

FEDERAL STATE BUDGETARY EDUCATIONAL INSTITUTION
OF HIGHER EDUCATION "BASHKIR STATE MEDICAL UNIVERSITY"
OF THE MINISTRY OF HEALTHCARE OF THE RUSSIAN FEDERATION
(FSBEI HE BSMU MOH Russia)

GENERAL PATHOPHYSIOLOGY

Manual for the practical classes



Ufa — 2018

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The study guide is prepared in accordance with the requirements of the Federal State Educational Standard of Higher Education end GEP in the specialty "General medicine" to study the discipline "Pathophysiology. Clinical pathophysiology " on the basis of the working program of the discipline and the current curriculum.

The manual sets out the modern purpose and learning objectives of each topic by the total of the pathophysiology, questions for preparation for classes, descriptions of experiments, situational tasks, a list of references.

The manual is intended for students in the direction of training "General medicine" (31.05.01), students using the intermediary language (English).

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FOREWORD

The main aim of this book is to help the students to understand the key notions of pathophysiology and to obtain maximum results during preparation for practical classes. The workshop presents a description of laboratory work with the necessary analysis of the results of the experiment and an explanation of the mechanisms of the observed changes. The use of situational tasks is aimed at the development of creative abilities of students, clinical thinking, necessary in the study of special disciplines in senior courses and in subsequent professional activities. This manual will enable to successfully fulfill the educational and control questions programme, as well as help to solve different medical problems, which may arise in future practical work and thus rich the ultimate goal of learning pathophysiology as a subject. The textbook can be used as a basic literature.

This manual is developed by the faculty of the Department of pathophysiology of the Bashkir state medical University in accordance with the General educational program and the requirements of the Federal State Educational Standard of Higher Education.

As a result of training, students master General cultural (GCC 1 – ability to abstract thinking, analysis, synthesis, GCC 5 – readiness for self-development, self-realization, self-education, use of creative potential), General professional (GPC 9 – ability to evaluate morphofunctional, physiological states and pathological processes in the human body to solve professional problems) and professional competences (PC 1 – ability and readiness to implement a set of measures aimed at the preservation and promotion of health and including the formation of a healthy lifestyle, prevention of the occurrence and (or) spread of diseases, their early diagnosis, identification of the causes and conditions of their occurrence and development, as well as aimed at eliminating the harmful effects on human health of its habitat, PC 6 – ability to determine the patient's main pathological conditions, symptoms, disease syndromes, nosolog-

ical forms in accordance with the international statistical classification of diseases and health-related problems, X revision, PC 21 – ability to participate in scientific research) (Federal State Educational Standard of Higher Education, 2016, working program in the discipline "Pathophysiology. Clinical pathophysiology", 2017).

The guide includes thematic laboratory classes that are built on a single principle and contains the following elements: educational goals to follow, assignments with a framework of activities, questions for testing and correcting the initial level of knowledge of students' skills, as well as for monitoring the mastery of the topic of the class, tasks for the development of creative abilities of students, clinical thinking, sample tests to control the final level of knowledge and a list of literature on all sections of the pathophysiology course.

This book consists of two parts. The first part of the "General Pathophysiology" includes material for conducting exercises on general pathophysiology and typical pathological processes. The second part of "Pathophysiology of organs and systems" contains material for studies devoted to pathology of organs and systems.

GENERAL PATHOPHYSIOLOGY. GENERAL NOSOLOGY. ETIOLOGY AND PATHOGENESIS

CLASS 1.

A SUBJECT AND METHODS OF PATHOPHYSIOLOGY. BASIC CONCEPTS OF GENERAL NOSOLOGY. INFLUENCE OF ENVIRONMENTAL FACTORS IN OCCURRENCE OF DISEASES (RADIANT ENERGY, OVERLOAD, ACCELERATION)

Learning objectives

Characterize objectives, main tasks, a method, and structure of pathophysiology as a science;

Identify main categories and concepts of general nosology and use them for the pathophysiological analysis of various forms of pathology.

Determine dependence of pathogenicity of environmental factors on experimental conditions.

Content

1. Knowledge control and correction.

Questions and discussion:

1. A subject, tasks, methods and structure of pathophysiology as a science.

2. Basic concepts and categories of general nosology: health, disease, pathogenic factor, typical pathological process, pathological state, pathological reaction.

3. Modeling: The main method of pathophysiology. Significance and possibilities of modeling the pathological processes and experimental therapy.

4. Limitations of modeling methods in experimental medicine and ways to overcome them.

5. Pathogenic effect of different types of radiant energy (solar spectrum, laser).

6. Types of ionizing radiation, the mechanism of action of ionizing radiation. Radiation sickness: Types, forms, stages, outcomes, and long-term consequences.

7. The concept of kinetosis. Types of acceleration, action of different accelerations.

8. The pathogenesis of disorders caused by accelerations, overloads, and weightlessness.

2. Assignments

Experiment № 1. Pathogenic effect of ultraviolet rays on unicellular organisms

Apply one drop of infusorial culture to each of three welled slides. Distribute droplets containing culture evenly over the entire surface of a well, and then check motility of infusoria under a microscope.

Take two slides, cover one of them with a Petri dish, and put both of these slides under a mercury quartz lamp at a distance of 50 cm from a light source. Hence, the third slide serves as a control. After 2-3 minutes of UV exposure compare infusorial motility in all three slides under a microscope. If there is no difference repeat the procedure for extra 1-3 minutes. Compare results and draw conclusions by answering the following questions:

1. Why do protozoa remain motile in a slide covered with Petri dish?
2. Which waves of UV radiation range have bactericidal effect?
3. What does cause the pathogenic effect of single-exposure excessive ultraviolet radiation (photochemical burn)?
4. What are the causes of mutagenic effect of prolonged excessive ultraviolet radiation?

Experiment № 2. Increased sensitivity of unicellular organisms to radiant energy under the influence of a photosensitizer.

Apply 1 drop of infusorial culture to 3 welled slides. Add 1 drop of tap water to the first slide and 1 drop of 0.0001% solution of the photosensitizer

(acridine orange dye) away from a quartz lamp and other light sources to the second and the third slides. Then place the first and the second slides simultaneously under a mercury-quartz lamp and cover them with a Petri dish. Put the third slide in a dark locker. 4-6 minutes after compare infusorial motility under a microscope in all 3 slides. Draw conclusions by answering the following questions:

1. Which compounds belong to the group of photosensitizers?
2. What is the mechanism of action of photosensitizers?
3. What is the mechanism of photoallergy in humans?

Experiment № 3. Investigation of vestibular and autonomic reactions in humans with Bárány Rotary Chair testing

2-3 volunteers of students may participate in this study. Check condition of a study participant including blood pressure, pulse, respiration rate, general condition, face' skin color, absence of spontaneous nystagmus, finger-nose-finger test, and writing test. Then a participant sits down in a rotary chair and undergoes 20 revolutions within 20-30 seconds; the acceleration under these conditions is 0.8 g.

After rotation, examine the participant again and compare findings with the initial results. Analyze the observed reactions and answer to the following questions:

1. What type of receptor does perceive acceleration?
2. What is the mechanism of vestibular and automatic disorders under the action of radial acceleration?
3. Why is vestibular disorder more pronounced in this experiment?

TASKS

Task 1.

Patient K., 43 years old, during the elimination of the accident at a nuclear power plant received a dose of 2000 BER (20 Gr). During the week, nausea, vomiting, bloating, bloody diarrhea, and fever progressed steadily. On the eighth day his condition worsened significantly. On examination: The condi-

tion is severe, t of the body is 39,8° C. Poor nutrition, dry skin, flabby, multiple pinpoint hemorrhages. Breath is hard, 40 per minute. Heart sounds are dull, irregular. Blood pressure 60/40 mm Hg. The abdomen is sharply swollen, intestinal noises are not heard. Absence of stool for 5 days.

Complete blood count: Hb – 80 g/l, erythrocytes – $2,5 \times 10^{12}/l$; leukocytes – $1,6 \times 10^9/l$; differential: basophils – no; eosinophils – no; myelocytes – no; bands – 12%; sticks – 23%; segmented – 70%; lymphocytes – 2%; monocytes – 1%. Platelets – $40,0 \times 10^9/l$. Urinalysis: color – reddish-brown, specific gravity – 028; protein — ++; glucose is negative. In sediment: erythrocytes in all fields of vision.

On the ninth day, death occurred in the presence of paralytic intestinal obstruction, shock, marked dehydration.

1. Name the pathology that developed in the patient.
2. Explain the mechanism of the main pathological processes (shock, dehydration, paralytic intestinal obstruction, sepsis).

Task 2.

One group of mice was irradiated with x-rays at a dose of 10 Gy for 10 minutes, the other at the same dose for 140 minutes.

In what case the damaging effect of radiation will be stronger and why?

- 1) in mice of the first group
- 2) in mice of the second group
- 3) the results are the same in both groups.

Task 3.

Technician K., 26, who was in the accident zone for a nuclear power plant for several hours, was urgently taken to the clinic complaining of nausea, vomiting, headache, increased body t, general weakness, drowsiness. During the following days of hospitalization, there were a further increase in body temperature, pulse lability, BP with a tendency to hypotension, meningeal signs, periodic changes of drowsiness and euphoria, growing confusion up to sopor. CBC shows neutrophilic leukocytosis and lymphocytopenia.

1. Name the pathology that developed in the patient.

2. Substantiate your conclusion.
3. Explain pathogenesis of these symptoms.

Task 4.

Patient A., 40, came to clinic complaining of general weakness, fatigue, sleep disturbance, headaches, decreased appetite, diarrhea. He considers himself sick for the last 2 years. The disease developed gradually, for no apparent reason. For the last 10 years he worked as a radiologist. Safety techniques were often neglected. Blood test: erythrocytes – $3,7 \times 10^{12}/l$, Hb – 85 g/l, reticulocytes – 0,1%, leukocytes $-3,8 \times 10^9/l$, lymphocytes – 14%. There are many hypersegmented neutrophils in the smear.

1. Name the pathology that developed in the patient.
2. Explain the mechanism of asthenic syndrome, changes in blood and digestion.

Task 5.

Physicist T., 35, was taken to the clinic 5 days ago with complaints about headache, dizziness, nausea, and vomiting. As a result of the accident, he was subjected to a general irradiation with gamma rays at a dose of 200 BER (2 Gr). At the moment he feels well, he does not make any complaints. CBC shows leukocytopenia, lymphocytopenia, decreased amount of reticulocytes.

1. What period of radiation sickness is present in the patient?
2. How would you explain patient's complaints at the admission?
3. What is the mechanism of changes in blood and what are the consequences of these changes?

Task 6.

Patient N., 32 years old, is treated at the clinic for acute radiation sickness. Two weeks ago, as a result of an accident, he was subjected to a general irradiation with a neutron flux (a dose of 500 BER or 5 Gr).

At admission, vomiting, diarrhea, headache, increased body temperature, erythema, blood pressure and heart rate were noted. After 2 days, he improved with only general weakness left. On day 14 the patient's condition worsened again. Body temperature rose to 39°C , pneumonia, stomatitis, and gingivitis were diag-

nosed. Point hemorrhages appeared on the skin and mucous membranes; blood revealed in sputum, urine, and feces. There were also oliguria and hyperazotemia. CBC: erythrocytes – $3,5 \times 10^{12}/l$; Hb – 70 g/l; there are no reticulocytes; leukocytes – $1,5 \times 10^9/l$, platelets – $25 \times 10^9/l$.

1. Explain the mechanism of hemorrhagic syndrome development.
2. Explain the mechanism of infectious complications and toxemia?

Task 7.

The patient K. was taken to a medical station. There is hyperemia of the face; pulse 130/min, blood pressure – 140/90 mm Hg. Respiration is frequent and superficial. Body temperature is 39°C. According to the accompanying witness, the victim, having liquidated the accident, worked for an hour at a temperature of about 60°C and high humidity.

1. What is the violation of the heat balance in this case?
2. Explain the mechanisms of increased heart rate with increasing body temperature.

Task 8.

The driver was in a car covered with snow for 24 hours. Pale skin, breathing and pulse were barely determined. Blood pressure – 70/40 mm Hg, body temperature (rectal) – 30°C.

1. What phase of hypothermia was observed in the patient?
2. What are the mechanisms of the described symptoms?

Task 9.

10-year-old boy after a long stay on the sun with uncovered head has general excitation, hyperemia of the face, frequent pulse, increased blood pressure and body temperature of 39°C.

1. What violation of thermoregulation did occur in the patient?
2. How does this violation of thermoregulation differ from hyperthermia?
3. What are the mechanisms of observed pathological phenomena?

Task 10.

A 16-year-old patient was taken to hospital in serious condition with a thermal burn of grade II (the burn area reaches 30% of the body surface). Con-

consciousness is confused, blood pressure – 80/50 mm Hg, pulse 120/min, weak filling, breathing frequent and superficial, body temperature – 37,7°C. Blood test: erythrocytes – $5,5 \times 10^{12}/l$, Hb – 170 g/l, hematocrit index – 0,52, leukocytes – $20 \times 10^9/l$.

1. What kind of shock did the patient develop?
2. What are the mechanisms of the revealed violations?
3. What is the mechanism of hemoconcentration in the patient?

Task 11.

Patient L., 9 years old, for a long time watched working electric welders. Arranged in the reception room with complaints about a burning sensation and "sand" in the eyes, sharply expressed photophobia, lacrimation. Examination reveals the conjunctiva of both eyes is hyperemic, edematous, pronounced lacrimation.

1. What is the name of this eye lesion?
2. What is it caused by?
3. What are the mechanisms of development?

Task 12.

Arriving home from the beach where Mikhail P., 18, spent 6 hours, he felt weak, dizzy, had throbbing headache, chills, nausea. 30 minutes later, he started vomiting; the temperature was 39°C. Aspirin and spasmalgin did not bring relief, the condition worsened even more, although body temperature dropped to 37°C, and Mikhail called an ambulance. On the way to a hospital, he lost consciousness, and he was taken to the intensive care unit.

1. What pathological process did develop in the patient? Substantiate the answer.
2. What are possible causes, stages and mechanisms of this process?
3. Why did the patient's condition continue to deteriorate despite the drop in body temperature?
4. What is the reason for the patient's loss of consciousness?

CLASS 2.

**CAUSAL RELATIONSHIP IN PATHOGENESIS. ACTION
OF CHANGES IN BAROMETRIC PRESSURE.
OXYGEN STARVATION**

Learning objectives:

Study damage manifestations at different levels of the organism's integrity.
Show unity of functional and structural changes in pathogenesis of disease.

Content:

1. Knowledge control and correction.

Questions and discussion:

1. Damage as the initial step of pathogenesis. Levels of damage: Submolecular, molecular, subcellular, cellular, organ, organism.
2. Structural, metabolic, physicochemical, and functional cell changes in reversible and irreversible damage.
3. Typical mechanisms of cell damage to genome, energy metabolism, fluid and solutes metabolism, protein metabolism, fat metabolism, carbohydrate metabolism, membranes and enzymes, receptors, intracellular repair processes. Role of free radicals and Ca^{2+} in cell damage.
4. Unity of functional and structural changes in pathogenesis of disease. Causal relationships in pathogenesis. Leading elements and vicious circles in pathogenesis.
5. Characteristics of the main types of hypoxia, causes, pathogenesis, manifestations. Role of occupational exposure in the development of hypoxic conditions.
6. Metabolic, morphological, and functional disorders in organism at acute hypoxic state. Hypoxia-induced sequence of disorders in the cell.

7. Adaptation to hypoxia: acute and long-terms types, conditions required for emergence of adaptation, mechanisms of development, manifestations. Features of hypoxia in infants.

8. Impact of low barometric pressure. Pathogenesis of mountain sickness and altitude sickness.

9. Impact of increased barometric pressure. Pathogenesis of caisson disease.

2. Assignments.

Experiment № 1. Influence of low atmospheric pressure and normal oxygen partial pressure.

Place rabbits in a pressure chamber. Create underpressure in the chamber corresponding to 3000 m altitude and supply oxygen to the level 2 times higher compared to the initial one. Every 2 km of "climb" note respiratory rate, presence/absence of cyanosis, behaviour and appearance of the rabbits.

1. What does maintain general condition of animals at normal oxygen partial pressure?

TASKS

Task 1.

A mouse is put in a small pressure chamber. Then, during 2-3 min the air is pumped out of it to decrease the pressure in the chamber down to 170-180 mm Hg. After 0,5-1 min in hypobaric conditions the mouse shows signs of anxiety: running over, scratching its snout. Two or three minutes later clonic and tonic seizures develop; the mouse is lying on its side, urinates: terminal breathing («gasping») develops. Soon afterwards, breathing stops, and the mouse dies. The overall time of the mouse's life in hypobaric atmosphere is 3-4 min.

1. What factors do affect the mouse in this experiment?

2. Which of these factors could be considered the leading cause of the pathological process — hypobaric hypoxia?

3. How can you experimentally test your presumptions about the nature

of the pathological process?

Task 2.

The air is pumped out of the pressure chamber to bring barometric pressure to approximately 20-30 mm Hg. Then the chamber is filled with pure O₂ up to the normal atmospheric pressure 750 mm Hg. A mouse is carefully put in the chamber so that the barometric pressure inside remains unchanged. The mouse is then observed as in the previous experiment. At first, seeking behavior is seen. Later the mouse is sitting quietly, showing no signs of pathology. Within approximately 10 min after the start of the experiment the mouse is taken out of the chamber, and its normal behavior and condition are established.

1. Taking into account the results of this test what can you say about the pathological factors which caused death of the mouse in the previous experiment?

Task 3.

The pressure chamber is filled with a mixture of gases composed of 95% N₂ and of 5% O₂. The barometric pressure inside the chamber is maintained equal to normal atmospheric pressure. A mouse is put in the chamber. A thin jet of the same gas mixture is being passed through the chamber. Partial pressure of O₂ in this gas mixture is about 37 mm Hg. After staying in the chamber for 10-15 min the mouse develops seizures and then dies.

Considering the results of all three experiments, answer the following questions:

1. Which factor caused the development of acute hypobaric hypoxia and death of the mouse?

2. What is the significance of hypobaria itself (decreased pressure of the inhaled air)?

Task 4.

Three mice are used in the following experiment. Mouse 1 is put in the water basin with water temperature 30°C. Five minutes later mouse 2 is put in the same basin. Ten seconds later both mice are taken out of the water and put

in the pressure chamber. The third mouse (intact) is also put in the same chamber. The air is pumped out as in experiment N 1 to bring the barometric pressure in the chamber down to 170-180 mm Hg. The intact mouse dies after 3-4 min of staying in hypobaric conditions. Mouse 2 dies after 6-7 min, and mouse 1 survives 15 min of hypoxia. At this point the experiment is stopped. When mouse 1 is taken out, no signs of pathology are observed.

1. What differences in resistance to hypoxia are seen in this experiment?
2. How can you explain the observed variations in sensitivity to hypobaric hypoxia?
3. Does hypothermia caused by evaporation of water from the surface of the body affect the resistance of mice 1 and 2 to hypobaric hypoxia? How can you name this factor in terms of pathophysiology?

Task 5.

In the study of various aspects of pathogenesis of hypoxia and edema a laboratory rat was intravenously injected a high dose of adrenaline. Immediately after the injection paws and ears of the rat turned pale; arterial blood pressure increased from 120/70 to 210/175 mm Hg; heart rate and respiration rate increased sharply. The level of PaO_2 remained unchanged but the levels of PvO_2 and PaCO_2 decreased. Nine minutes after the adrenaline injection despite hyperventilation the rat developed acrocyanosis. The content of gases in the arterial blood remained normal, but a progressive decrease in the PvO_2 level was noted. Four minutes later dyspnea and moist rales in the lungs developed; blood pressure decreased sharply; pulse pressure dropped, and heart beats became irregular. At the same time PaO_2 started to decrease, and PaCO_2 increased. By the end of the 18th minute the rat developed clonic and tonic seizures, agonal breathing (gasping) and foamy, reddish discharge from the respiratory tract. Soon it died.

1. Can we state that despite activation of the heart function, increased vascular tone, and a high blood pressure the rat suffered from hypoxia immediately after the adrenaline injection? If you think so, substantiate your opinion. If

you do not agree, name the time point by which hypoxia has developed.

2. What are the type(s) and pathogenesis of such hypoxia?

3. Did the rat develop edema? If you think so, name the organs where it occurred and the time point after the adrenaline infusion when edema became evident?

4. If, in your opinion, edema developed in the rat, what forms of pathology in humans may lead to edema with similar localization?

Task 6.

An accident occurred with the aircraft during its flight at the height of 10.000 m. The cabin of the aircraft lost its hermetic state, and in several seconds the barometric pressure inside fell down to atmospheric level of 170 mm Hg at this altitude. The crew failed to make an urgent lowering maneuver, and the aircraft continued the flight at the same height for several minutes.

1. What pathological processes developed in the body of passengers during the aircraft accident? Which of them were the most dangerous for their well-being and can lead to death?

2. What are the likely causes and mechanisms of these pathological processes?

3. What types of preexisting pathology could include the highest risk for life in this situation?

Task 7.

A 30-year-old woman had been brought to hospital by ambulance as emergency case with a preliminary diagnosis of rupture of an uterine tube with internal bleeding due to ectopic pregnancy. On the way to hospital the patient lost consciousness several times. Examination in hospital showed that the patient was conscious, but confused. She was markedly pale, had a weak, rapid pulse; auscultation revealed muffled heart sounds; blood pressure 80/35 mm Hg. Examination showed the presence of varicose veins in the extremities. The patient told the physician that her condition worsened suddenly when she was eliminating hair on her legs with a depilator. Unexpectedly she felt shortness of

breath, palpitations, and pain in the heart. Later on in hospital, despite inhalation of oxygen, she went on complaining of shortness of breath.

1. Which of the typical pathological processes and what type of it could develop in the patient? Substantiate your answer.

2. Name the symptoms caused by this pathological process. What are the mechanisms underlying each of these symptoms?

3. Describe the causative factors of this pathological process. Which of them are most likely to occur?

4. Why was the inhalation of oxygen ineffective to alleviate shortness of breath?

Task 8.

When a physician of the emergency service arrived at the site of the vehicle accident he found that the victim of the accident developed shock associated with an open fracture of the right leg and massive blood loss. The patient also had a closed chest trauma. When bleeding was stopped by application of a tourniquet, the physician injected the patient subcutaneously an appropriate dose of morphine (to alleviate pain) and a stimulant of the cardiac function. However, the treatment had no effect, and the symptoms of shock were progressing. Pulse beats were detectable only in the large arteries (60 beats/ min), blood pressure decreased to 60/25 mm Hg. On auscultation breathing was depressed over the left chest, and there were no breathing sounds over the right chest. Several minutes later respiration became shallow and ineffective: the patient started to «gasp air». To stimulate breathing the physician injected subcutaneously a stimulant of the respiratory center. However, this treatment also had no effect.

1. What type (or types) of the emergency condition developed in the patient?

2. What types of hypoxia and in what order developed in the patient?

3. Describe the possible causes and pathogenesis of various types of hypoxia.

4. Explain the typical changes in pH, PaCO₂, SaO₂, SvO_a, blood oxygen capacity, the minute volume of breathing, and the minute volume of circulation during these types of hypoxia.

5. Why was the treatment provided by the physician ineffective?

Task 9.

A 30-year-old patient K. had been treated in the intensive care unit after amputation of uterus which was done under an endotracheal ether narcosis. Suddenly the patient's condition worsened sharply. She developed choking sensation, dyspnea, chills, psychomotor retardation and apathy. Her skin turned pale; acrocyanosis became evident. Physical examination showed an increased respiration rate of 28 per min, muffled heart sounds, no rales during auscultation of the chest; the pulse was regular 120 beats/min; blood pressure 65/30 mm Hg; concentration of Hb 100 g/l; hematocrit 0,30 (N:0,36-0,42). The patient was intubated and supplied by oxygen but this measure had no effect.

1. What pathological processes developed in the patient after the surgery? Substantiate your answer.

2. Can the worsening of the patient's condition be associated with the postsurgery development of hypoxia of: respiratory type, circulatory type, anemic type, histotoxic type?

3. What are the possible causes and mechanisms of development of each of the presumed types of hypoxia in this case?

4. Are there any signs of the urgent adaptive mechanisms that could compensate hypoxia in this patient? Describe them if they are present. Why do they appear to be ineffective in this case?

Task 10.

A 50-year-old patient K. had undergone gastroectomy under narcosis with the use of mechanical ventilation. The surgery was performed urgently due to extensive stomach bleeding which developed suddenly as a result of erosion of the large blood vessel in the decaying tumor. During the treatment of shock and in the course of the surgery the patient received infusions of volume

expanders (approximately 1 L) and transfusion of 2,5 L of whole donor blood which had been stored for 2 days. By the 3rd day after the surgery, despite the recovery of the normal blood values of Hb, the patient complained about weakness, headache, dizziness. On examination: the skin of the extremities was cold; severe dyspnea and renal insufficiency were observed; in the conjunctiva a tint of jaundice appeared. Eventually, mechanical ventilation was initiated to treat the patient.

1. What pathological processes developed in the patient after the surgery? Substantiate your answer.

2. What are the possible causes and mechanisms of development of each of the pathological processes: before the surgery; in the course of the surgery and immediately after its completion; by the 3rd day after the surgery?

3. What adaptive reactions could be activated in the patient as a result of hypoxia? Why were these adaptive reactions ineffective in this case?

Task 11.

A 60-year-old patient C. had an extensive transmural infarction in the anterior wall of the left ventricle two weeks before. During his stay in hospital he woke up at night with choking sensation and called for help. When a nurse came she helped the patient to sit up on the bed with his feet down on the floor. Then she opened the window. The patient felt a little better. However, 10 min later he began complaining of shortness of breath, the need coughing regularly (without secretions), and audible rales in the chest. The nurse gave the patient supplemental oxygen by face mask and called a physician.

1. What types of hypoxia did the patient suffer from? Substantiate your answer.

2. In what succession did these types of hypoxia develop?

3. How and why did the parameters PaO_2 , $PaCO_2$, PvO_2 , $PvCO_2$, $SaCO_2$, SvO_2 , pH change in the patient's blood? Substantiate your answer.

4. What are the possible mechanisms of immediate adaptation to hypoxia in patient C.?

Task 12.

At an altitude of 10,000 m there was an emergent decompression of the passenger aircraft. Within a few seconds the barometric pressure in the cabin became equal to the atmospheric pressure at a given altitude (170 mm Hg). The crew could not make an urgent descent, the plane continued flying at that altitude for a few more minutes.

1. What pathological processes that have occurred in the body in passengers in connection with this emergency situation, can threaten their health and lead to death?
2. What are the causes and mechanisms of development of these pathological processes?
3. Which of the processes you named represents the greatest danger to the lives of passengers and why?

Task 13.

Pilot V., 40, entered the hospital to determine the fitness for flight. To this end, was tested in a pressure chamber. Five minutes after the "rise" to a height of 5000 meters he began to complain about a headache and dizziness. There were also shortness of breath, cyanosis of fingertips, face turned pale. Blood pressure increased from 120/70 to 130/75 mm Hg, pulse 120 per min, weak filling. Suddenly, the pulse and breathing began to decrease, the blood pressure decreased, the subject lost consciousness.

1. Name the developed pathological process.
2. Indicate its main pathogenetic factor.
3. Name and justify the measures to help the victim.

CLASS 3.

CONTROL QUESTIONS

1. Pathophysiology as a subject. Tasks of pathophysiology as “philosophy of medicine”. Interrelations of pathophysiology with other medical disciplines. Value of pathophysiology for contemporary clinical and prophylactic medicine. Ecological aspects of pathophysiology.

2. Basic stages in development of pathophysiology (Hippocrat, Halen, Demokrit, Wirchow, Selie). Humoral and solidary theories of pathology. Iatrophysical and iatrochemical referrals in medicine.

3. Basic stages in development of Russian pathophysiology, its peculiarities, principles (evolutionary – Mechnikov, nervism – Sechenov, Botkin, Pavlov). Methods of pathophysiology, experimental simulation of diseases.

4. Common doctrine about disease (Pashutin, Ado, Zaiko etc.). Conception of “health” and “disease”. Periods of disease and its clinical outcomes. Iatrogenic diseases and deontological aspects of pathophysiology and medicine.

5. Terminal conditions. Fundamental regularities of different stages of dying (decreasing in regulation of CNS, cor, breathing, metabolism etc.). Clinical and biological death. Common principles and methods of resuscitation (Negovskii).

6. Role of causes and environment condition in genesis of diseases. Analysis of some conceptions of fundamental etiology (monocausality, constitutionalism, theory of factors etc.).

7. Conception of pathogenesis. Comprehension of local and total, leading sections of pathogenesis. Positive and negative biofeedback. Primary and secondary damages.

8. Sanogenesis. Compensatory functions of the organism and their role in mechanisms of recovery. Role of nervous, endocrine, and immune systems in recovery.

9. Kinetoses: definition, causes, pathogenesis. Acceleration, overloads,

imponderability (disorders caused by this conditions). Disorders caused by space travel.

10. Pathological processes linked with mechanical factors – distension, rupture, crush syndrome.

11. Pathological action of high environment temperature on the organism. Pathogenesis of heat and sun strokes.

12. Pathological action of low temperature on the organism. Common cooling, frostbite, catarrhal diseases. Application of artificial hypothermy and hibernation in therapeutic aims.

13. Pathological action of infrared, ultraviolet, and laser rays on the organism.

14. Damaging action of ionizing radiation on the organism. Radiation sickness: classification, stages of development, pathogenesis and long-term consequences. Ecological aspects of this issue.

15. Ill effect of the electric current to the organism (pathogenesis). Mechanisms of disorders in functions of organism. Causes of death from electrotrauma. First aid.

16. Ill effect of the acoustical energy on the organism (sound, ultrasound). Early and long-term consequences of noise action to the organism in industrial conditions. Use of ultrasound in medicine.

17. Action of low barometric pressure on the organism. Mountain and high-altitude diseases, stages of development. Principles of prophylaxis and first aid.

18. Action of increased barometric pressure on the organism. Pathogenesis of caisson disease. Principles of prophylaxis and therapy.

19. Characteristics of the main types of hypoxia. Causes of hypoxia. Pathogenesis.

20. Mechanisms of urgent and long-term adaptation to hypoxia. Structural and functional damage of organs and tissues during hypoxia. Different stability of tissues and organs to hypoxia.

21. The concept of stress as a nonspecific reaction of the organism to extreme effects. Stages and mechanisms of stress development. Protective-adaptive and pathogenic importance of stress. The concept of "adaptation diseases". Criticism of some provisions of the concept of stress G. Selye.

22. Cell damage, causes, types, stages (paranecrosis, necrobiosis). Specific and nonspecific forms of damage. Apoptosis, its significance in norm and pathology.

23. Common mechanisms of cell damage. Damage to membranes and cell enzymes, the importance of lipid peroxidation. The imbalance of sodium, potassium, calcium ions and liquid in the mechanisms of cell damage.

24. Violation of the structure and function of subcellular structures, biological membranes.

25. Mechanisms of protection and adaptation of cells in case of damage. Adaptive changes in the functional activity of cells. Cellular and subcellular regeneration.

CLASS 4.

REACTIVITY AND RESISTANCE OF ORGANISM, ROLE IN PATHOLOGY. FACTORS AFFECTING THE REACTIVITY AND RESISTANCE

Learning objectives: study the role of reactivity and resistance in the origin and course of pathological processes.

Content:

1. Knowledge control and correction.

Questions and discussion:

1. The concept of reactivity and resistance, similarities and differences. Impact of reactivity in the development of pathological processes.
2. Types, forms, and mechanisms of reactivity and its determining factors. Features of reactivity in childhood.
3. Pathology of nonspecific defensive factors (phagocytosis, complement, properdin): causes, mechanisms of development, manifestations. The role of nonspecific factors in the mechanisms of protection of the oral mucosa.
4. Body constitution: Definition, role in reactivity, classification of constitutional types.
5. The role of body constitution in the emergence and development of diseases. Features of physiological and pathological processes in people of different constitutional types.
6. Diathesis: Types, significance in pathology (lymphatic-hypoplastic, exudative, neuro-arthritis).
7. The role of heredity in reactivity and resistance. Methods of studying heredity. Similarities, differences and origin of acquired, congenital, hereditary and family forms of pathology. Importance of critical periods in the pathology of embryo and fetus.
8. Genomic and chromosomal diseases. Mutations as an initial step in the evolution of hereditary information. The importance of environmental protection. Possible ways of preventing hereditary diseases.

9. The concept of hereditary and congenital disorders, phenocopies.
Classification of genetic disorders.

10. Classification of mutations and mutagens.

11. Mendelian disorders: definition and patterns of inheritance.

12. Codominance, pleiotropism, genetic heterogeneity.

13. Definition of penetrance and expressivity. Factors determining penetrance and expressivity.

14. Features of autosome-dominant disorders. Examples of disorders transmitted by autosome-dominant trait.

15. Features of autosome-recessive disorders. Examples of disorders transmitted by autosome-recessive trait.

16. Features of sex chromosome-linked inheritance. Examples of disorders transmitted by X-linked trait.

17. Biochemical and molecular basis of single-gene (Mendelian) disorders.

18. Features of multifactorial (polygenic) disorders. The most common multifactorial disorders.

19. The concept of chromosome disorders, their mechanism. Mosaicism. Types of chromosome aberrations: deletion, balanced and Robertsonian translocation, ring chromosome, inversion.

20. Examples of the most common chromosome disorders. Monosomies and trisomies.

2. Assignments.

Experiment № 1. Manifestation of individual reactivity in mice with intraperitoneal injection of foreign protein.

You need 4-6 mice for this experiment. The reactivity index is a change in number of peripheral blood leucocytes, which occurs in response to intraperitoneal administration of the same dose of foreign protein (milk, serum, vaccine).

Put a tail of each mouse in a jar with warm water for 2-3 minutes to improve blood circulation, then wipe it dry with a napkin, cut the skin at the tip of

a tail (no more than 1 mm) with scissors and take blood samples into the mixer for the white blood cells counting.

Make intraperitoneal injection of 0.2 ml milk to each mouse immediately after taking blood samples. Further, obtain blood samples again in 30, 60 and 90 minutes after the injection.

Count numbers of white blood cells in all the samples, draw the curves of changes in the. Analyze the curves, write down the conclusions, and answer the following questions:

1. What forms of reactivity did you observe in the experiment?
2. Do these manifestations refer to protective or pathological reactions?

Why?

3. Describe organism predisposition to diseases due to individual features of reactivity.

Leucocyte counting.

A. Take a blood sample with leucocyte pipette up to '0.5' mark and dilute it 20 times up to the '11' mark with a 3% acetic acid coloured with methylene blue. Count leucocytes in Goriaev' chamber (haemocytometer) in 100 large squares of the grid.

B. Add 0.4 ml of 3% acetic acid and 0.02 ml of blood taken with a pipette from the Sahli hemometer to Vidal tube. Shake the tube with diluted blood thoroughly before filling the Goriaev' chamber (haemocytometer).

Use the following formula to calculate the number of leucocytes in 1 litre:

$$X = \frac{L \cdot 4000 \cdot 20}{16 \cdot 100} \cdot 10^6,$$

L – Number of counted leucocytes;

16 x 100 – the number of small squares in 100 large ones;

20 – Blood dilution;

1/4000 – the volume of 1 small square = 1/20 x 1/20 x 1/10 = 1/4000 mm³, where 1/20 is a side of the square, 1/10 is the depth of the square under the cover glass.

Experiment № 2. Leucocyte recruitment to the oral cavity by chemotaxins.

Experiment consists of the 3 stages:

1. Estimation of the basal leucocytes amount by oral flush.

Clean the mouth cavity by thoroughly rinsing it with tap water for 2 minutes. Use the glass № 1 for this.

In 10 minutes put 10 ml of saline with a syringe into the glass № 1 and rinse the anterior section of the oral cavity for 2 minutes. Collect flush in the glass № 1 and put it aside.

2. Taking oral flushes after rinsing mouth with fresh meat broth.

Rinse the front of mouth for 2 minutes with 5 ml of fresh beef broth. After rinsing spit the broth into the sink. In 10 minutes after the broth exposure rinse the front section of mouth with saline for 2 minutes and collect flush in a glass № 2. After 10 minutes, repeat the procedure and collect flush in a glass №3.

3. Leucocyte counting.

Mix liquid in the glass № 1 thoroughly with pipetting it 7-10 times.

Transfer the mixed liquid with the same pipette to a clean weighting bottle up to '0.5' mark. Dry the pipette with gauze. Collect 3% acetic acid with the same pipette up to the '1.0' label and transfer it to the weighting bottle with liquid. Stir the mixture with a pipette and leave it for 5 minutes.

Prepare Goriaev' chamber (haemocytometer) to count the blood cells.

Fill the chamber with a mixture from glass № 1 and put it under a microscope. Use eyepiece 15 and lens 8 to count white blood cells in 100 large squares (Figure 1).

Attention! You may also find squamous epithelium cells and food remnants that should not be counted.

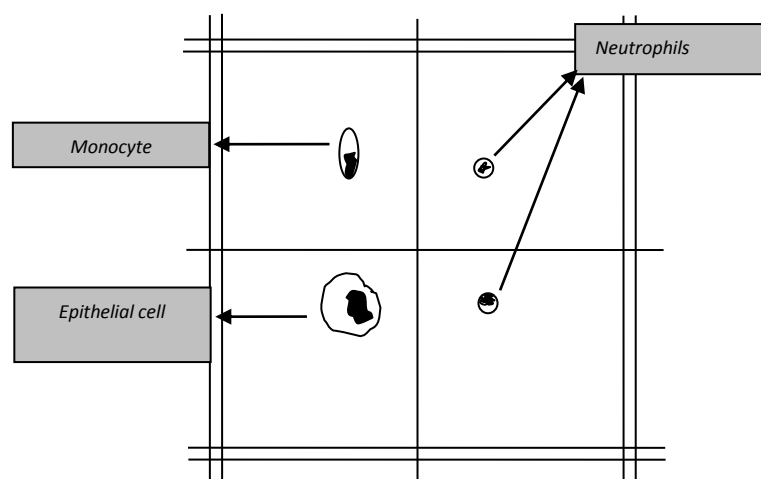


Figure 1.

Count leucocytes in 1 μ l according to formula:

$$\text{Number of leucocytes} = \frac{a \times 4000 \times 3}{1600} = a \times 7,5$$

1600 – Number of small squares in 100 large squares in the Goriaev' chamber;

1/4000 – the volume of 1 small square in mm;

a – number of counted leucocytes.

Calculate leucocyte count in flushes № 2 and № 3 using the described formula.

Determine the leucocyte migration index (LMI) for samples № 2 and № 3 according to the formula:

$$\frac{L_x - L_1}{L_1} \times 100 \%$$

L_1 is the number of leucocytes in the background sample;

L_x is the number of leucocytes after rinsing with broth (glass № 2 and № 3)

Table 1

Influence of meat broth on leucocyte migration

	Sample number	Background	After rinsing with broth	
			In 10 minutes	In 20 minutes
Number of leucocytes in 1 μ l of the sample				
LMI (leucocyte migration index)				

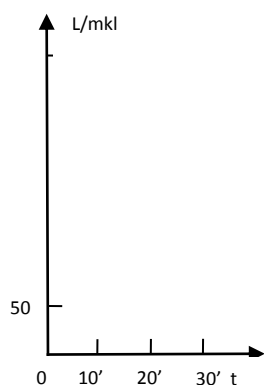


Figure 2. Leucocyte migration under the influence of meat broth on the mucous membrane of the oral cavity

Discuss the results and draw conclusions by answering the questions:

1. What kind of reactivity does the observed change belong to?
2. What is the mechanism of the observed reactions?
3. What form of individual reactivity (normal, hyperergic, hypoergic) did you observe?

Experiment № 2. Viewing a stained smear (demonstration sample).

Smears were prepared in advance from the supernatant obtained after centrifugation of the flushes from glass № 2.

Examine the smear under a microscope with immersion. Describe the morphological composition of the flush. Draw leucocytes in different stages of degeneration, epithelial cells, and microorganisms.

TASKS

Task 1.

N., a healthy woman, visited a genetic counseling unit for consultation. She told her doctor that her father suffered from color blindness, but her mother had no vision problems. N. was anxious about the risk of color blindness to her future children.

1. What are the manifestations of color blindness and the pattern of its inheritance?
2. Can N. carry the gene of color blindness in her genotype?
3. If she can, what is the probability of disease and the probability that her children (separately boys or girls) will be the carriers of the abnormal gene?

Task 2.

N., a healthy woman, whose father suffered from hemophilia A and mother was healthy, went to a genetic counseling unit. She was anxious about the risk of hemophilia to her grandsons. Her husband, as well as her son and two daughters were healthy.

1. What is the pattern of inheritance of hemophilia A? What are the main features of this type of inheritance?
2. What is the risk of the disease descending from her son to her grandson and from her daughter to the grandchildren?
3. What are the etiology and pathogenesis of hemophilia A?
4. Can this disease have lethal and sublethal forms?

Task 3.

Patient S., a pregnant woman, went to a genetic counseling unit for consultation. She told her doctor that her sister suffered from phenylketonuria. In her husband's pedigree there are marriages between close relatives, but none of the children have had phenylketonuria. A thorough examination of the patient and her husband revealed no signs of pathology.

1. How great is the risk of phenylketonuria to patient S.'s sons?
2. What are the etiology and basic mechanisms of phenylketonuria? Is gender significant for its inheritance?
3. What are the main manifestations of the disease and their pathogenesis?
4. What is the approach to early diagnostics of phenylketonuria in the newborns?
5. Is it possible to prevent phenylpyruvate oligophrenia in children?

Task 4.

Patient M., male, 21 years old, suffers from hereditary form of hypogammaglobulinemia. His father also suffers from this disease; mother is healthy. An examination showed a significant decrease of IgM and IgG levels in the patient's blood.

1. What are the etiology, basic mechanisms and clinical manifestations of hypogammaglobulinemia?

2. What is the pattern of its inheritance?

3. Characterize the genotype of the patient's mother with respect to this pathology, and also the possible genotypes of the patient's siblings. How much is the risk for the patient's sibs to develop hypogammaglobulinemia?

4. Does the pattern of immune and allergic reactions change in hypogammaglobulinemia? Substantiate your opinion.

Task 5.

In a genetic counseling unit female patient Z. told the doctor that her sister suffered from severe sickle cell disease, but she and her husband were healthy. Z. was concerned about the degree of risk of sickle cell disease to her future children. To answer this question the doctor investigated the types of Hb in the patient's and her husband's blood. The test revealed that in the patient's erythrocytes there were 70% of HbA and 28% of HbS; in her husband's erythrocytes there were only 98% of HbA and no HbS.

1. What is the pattern of inheritance of sickle cell disease?

2. What are the chances for the patient's children to develop sickle cell disease? What is the probability that her children would be healthy carriers of the abnormal HbS gene?

3. Does the development of sickle cell disease depend on the sex?

4. What factors can aggravate the course of the disease?

Task 6.

Population's studies of the frequency of various diseases in the city of N. among monozygotic (MZ) and dizygotic (DZ) sibs revealed the following:

Table 2

Diseases	Frequency in MZ, %	Frequency in DZ, %
a) Schizophrenia	87	4
b) Scarlet fever	95	94
c) Poliomyelitis	44	39

1. What is the value of the Holtzinger's factor for these diseases?

2. Estimate the role of genetic and environmental factors in the development of these diseases.

Task 7.

A male patient, 3 years old, was admitted to the pediatric department. On examination: signs of growth retardation, flat facial profile, half-open mouth, oblique palpebral fissures and epicanthic folds are evident; transverse skin folds on the palms. A study of the patient's karyotype revealed the following: 46, XY, + t (+14, 21).

1. What disease does the patient suffer from?
2. Describe and characterize the patient's karyotype? By what features does it differ from the normal one?
3. What are the possible causes and pathogenesis of this disease?
4. Are other variants of karyotype modification possible in this disease? Which of the variants is the most common?

Task 8.

During the epidemic of flu, the local doctor noticed that the patient suffering from hyperthyroidism had hyperergic disease, and the patient with hypothyroidism had a hyperthyroid.

1. Give an explanation of these observations.
2. Is the hyperergic course of the disease always accompanied by a high resistance of the body?

Task 9.

Two rats: one intact, the other after bilateral adrenalectomy (adrenalectomy) is placed in a large jar with water and watch how the animals swim. After 10-15 minutes the adrenalectomized rat begins to sink and it is removed, and the other continues to float for a long time.

1. Assess the reactivity and resistance of these animals in relation to physical activity.
2. Explain the results.

Task 10.

Three rabbits: the first – intact, the second – after the previously injected subcutaneously with caffeine and the third – in a state of narcotic sleep (hexsenal), a 1% solution of pyrogenal was injected subcutaneously at the rate of 1

ml / kg of body weight. After 2 hours, the rectal temperature was measured. Results: in the first rabbit, the temperature rose by 1°C, in the second rabbit by 3°C and in the third by 0,2 °C.

1. Evaluate the reactivity form of these animals for the introduction of a pyrogenic substance.

2. What kinds of reactivity are these changes?

3. Explain the results.

Task 11.

The control group of animals (white rats) and the experimental one, who consumed low-calorie food with a limited content of vitamins, were subjected to a long load on the treadmill. The running time in the control was 45 minutes, the blood sugar level increased 1,5 times, the concentration of pyruvate increased by 20%. Running time in the experimental group of animals was 20 minutes. against a background of hypoglycemia, the concentration of pyruvate increased to 180% in relation to the norm.

1. What factors have determined such reactivity of animals?

Task 12.

Control and experimental (after thyroidectomy) groups of animals were placed in a chamber with a reduced oxygen content. After 5 days, the control animals died, and the experimental one showed a decrease in metabolism, hypoproteinemia, inhibition of reflexes, and a decrease in the function of the organism.

1. What factors determined such reactivity of control and experimental animals?

CLASS 5.

IMMUNE REACTIVITY. IMMUNOPATHOLOGY

Learning objectives: conduct a pathophysiological analysis of clinical and model situations associated with the development of immunopathological states and reactions.

Contents:

1. Knowledge control and correction.

Questions and discussion:

1. System of immune surveillance, its structure, functions, role in organism. Mechanisms of immunity, specific and natural protective factors.

2. Typical forms of impaired immune reactivity: Classification, general characteristics.

3. Establishment of immune reactivity in early childhood

4. The role of immune mechanisms in dental diseases.

5. Primary B-cell (humoral) immunodeficiencies: Causes, mechanisms of development, manifestations.

6. Primary T-cell (cellular) immunodeficiencies: Causes, mechanisms of development, manifestations.

7. Primary combined immunodeficiencies: Causes, mechanisms of development, manifestations.

8. Secondary (acquired) immunodeficiency and immunodepressive conditions, causes, principles of treatment.

9. Acquired Immunodeficiency Syndrome (AIDS). Etiology, epidemiology, pathogenesis, consequences.

10. Pathologic immune tolerance: Causes, manifestations, consequences.

11. Graft-versus-host disease: Causes, types, mechanisms of development.

2. Assignments.

Experiment № 1. Influence of epinephrine and acetylcholine on the phagocytic activity of blood leucocytes.

Add in series 1 part of heparin, 2 parts of blood, 1 part of flush with 24-hours Staphylococcus culture in 3 numbered tube, whereas 1 part corresponds to 30 divisions of the Panchenkov' capillary. Use 3 different pipettes to collect heparin, blood, and Staphylococcus culture.

Mix thoroughly the tubes, add 2 drops of epinephrine ($1:10^8$ dilution) to the 1st tube, 2 drops of acetylcholine ($1:10^5$ dilution) to the 2nd tube, and 2 drops of saline in the 3rd tube (control).

Place all three tubes simultaneously in a thermostat at 37°C. Expose for 15 minutes, mix the tubes again, and make smears on slides from each tube with indication the number of a tube on a slide. Dry the smears in the air, fix them with a Nikiforov' mix for 30 minutes or with methanol for 3 minutes. Stain the smears for 30 minutes with Romanovsky-Giemsa method. Wash paint off with tap water and dry the smears in the air.

Calculate for each slide the following parameters using the immersion microscopy:

- 1) phagocytic index;
- 2) phagocytic number.

To do so count 100 neutrophils and monocytes, including phagocytes with and without microbes. Mark active phagocytes that absorbed microbes by number corresponding to the number of cocci within phagocyte; denote intact phagocytes by zeros.

Phagocytic index indicates an average number of microbes per 1 phagocyte. It could be calculated as amount of absorbed staphylococci divided by amount of all counted phagocytes (100).

Phagocytic number equals to number of phagocytes that have absorbed microbes out of 100 counted phagocytes. Hence, phagocytic number is the percentage of active phagocytes.

Note the change in the indices of phagocytic activity after exposure to adrenaline or acetylcholine draw conclusions.

1. What stages of phagocytosis did you observe?
2. How and why did phagocytic indicators change under the influence of acetylcholine and adrenaline?
3. What bactericidal mechanisms do phagocytes activate during the capture of microorganisms?
4. What is the clinical significance of leucocytes' phagocytic activity?

TASKS

Task 1.

A patient who had undergone a surgical intervention for gallstones presented with a slow healing of the postoperative wound, its infection, and fever. The use of antibiotics (after sensitivity tests) had positive effect Investigation of the patient's blood showed leukopenia due to a decreased neutrophils and monocytes, a decreased mobility of leukocytes and a low activity of leukocytic myeloperoxidase. Similar abnormalities were found in the patient's sister and brother.

1. Taking into account the observed changes in the leukocyte counts, name the syndrome(s) which the patient suffers from. Substantiate your answer.
2. What are the possible causes, manifestations, and mechanisms of this syndrome? Are there grounds for its hereditary or congenital origin?
3. What are the mechanisms of the decreased neutrophil mobility and low activity of the neutrophil myeloperoxidase?
4. What approaches can be used to increase the antimicrobial resistance of the patient's body?

Task 2.

A 20-year-old patient who suffered from diabetes mellitus presented with slow healing of a surgical wound, purulence, and an increase in body temperature up to 37,2°C after he had undergone the appendectomy. Treatment with antibiotics for 6 days after the appropriate sensitivity check had no effect. With

an aim to find the reason for treatment failure blood glucose determination and special neutrophil tests were performed. The results of the tests are the following: hyperglycemia (320 mg/dl) is accompanied by decrease in the number, mobility, and microbicidal activity of neutrophils.

1. What parameters, in your opinion, should be investigated to elucidate mechanisms of the decreased mobility and microbicidal activity neutrophils?

2. What are the most likely causes and mechanisms of impairment of the neutrophils' function in this patient?

3. Is there any association between the abnormal phagocytosis, the infection of wound and its slow healing? If you think there is, name and characterize possible mechanisms of such an association.

4. Why is the body temperature in the patient with a purulent wound increased only moderately?

5. Why did the treatment with antibiotics in this case have no effect?

Task 3.

During examination of an 11-year-old patient with albinism of the skin and irides a physician found multiple skin blisters filled with pus and multiple petechiae. Blood tests revealed the presence of marked anemia, thrombocytopenia, and neutropenia. Special investigation of neutrophils showed a decrease in mobility and microbicidal activity, and also the presence of large granules in the cytoplasm. Hyporesponsiveness of NK-cells was also detected.

1. What type of pathology does the patient suffer from?

2. What are the likely causes of abnormal mobility and microbicidal activity of neutrophils in this patient?

3. What is the cause of this pathology and what are the mechanisms of its main manifestations: anemia, thrombocytopenia, and leukopenia?

Task 3.

A 28-year-old patient K. visited his physician with complaints about recurrent stomatitis, tonsillitis, tracheitis, otitis, and repeated pneumonia even in the summer time. Procedures aimed at enhancing the nonspecific body resistance to infection, such as an increase of cold endurance, had no effect. The

results of the laboratory tests are the following: lymphocytes reactivity toward phytohemagglutinine (PHA) and tuberculin is normal; activity of the complement factors and levels of IgM, IgG, and IgA in the blood serum are within the normal range. The complete blood count test with differential showed no changes in erythrocytes count or Hb concentration; leukocytes count is low due to a decrease in monocytes, but the content of granulocytes is normal. The phagocytic activity of macrophages is decreased by 45%.

1. Define the type of pathology found in the patient.
2. What part of the immune surveillance system is impaired in the given patient: the specific immune response or the nonspecific host defense? Substantiate your answer.
3. If the specific immunity is abnormal, what part of it is defective: A, B or T? Explain the origin of symptoms presented by the patient.
4. If the host defense system is deficient, what subsystem is responsible of the defect? Explain the mechanisms of symptoms seen in the patient.

Task 4.

An 8-year-old patient was admitted to the pediatrics department, parents reported frequent occurrence of otitis, quinsy, rhinitis, conjunctivitis, bronchitis, pneumonia, and enterocolitis in their child. The current admission was associated with a high risk of bacterial endocarditis and sepsis in the young patient.

A test performed in the hospital showed the presence of leucopenia due to a decrease in lymphocytes, especially in T-cells, and, to a less degree, in B-cells, a decrease in the blood levels of IgA and IgE (by 50 % respectively, compared to normal values); concentration of IgG at the lowest normal value; proliferation of lymphocytes in the presence of phytohemagglutinine is decreased.

1. Define the type of pathology developed in the patient. Substantiate answer.
2. What are the possible causes of this pathological state?
3. Taking into account the results of the laboratory tests, explain the the-

ory of the development and consequences of this type of pathology.

4. How can you explain a decrease in the lymphocytes' reaction phyto-hemagglutinine and a sustained reduction in the blood of IgA and IgE when the levels of IgG are in the normal range?

5. What are the symptoms of this IgA and depletion?

Task 5.

A 32-year-old male patient visited his dermatologist with complains about weakness, rigor, long-lasting sustained fever, and night sweating. About a week ago he noticed the emergence of pink-violaceous spots and nodules on his right shin with a tendency to expand and coalesce. The use of antihistamine drugs and glucocorticoids had no effect on the skin lesions.

On examination: the similar lesions were found in the mouth mucosa and conjunctiva; the subclavian, supraclavicular, submandibular, preauricular, retroauricular, and tonsillar lymph nodes were enlarged. Complete blood: count test showed no changes in erythrocytes count, and the presence of leukopenia due to a decrease in lymphocytes and monocytes.

Over 3-year period before the onset of the disease the patient had been working in Africa where he experienced pneumonia three times during the last year of his business trip.

1. What pathological processes developed in the patient? Substantiate your answer.

2. If you see several pathological processes, which of them is primary and which are secondary? Explain their interrelationship.

3. What is the cause, and what are the main stages of development of the primary pathological process? Substantiate your opinion using the given data.

4. What are the mechanisms of development of leukopenia, lymphopenia, and monocytopenia? What type of lymphocytes, T or B, contributes more to the development of lymphopenia? Substantiate your answer.

Task 6.

After spending an hour and a half in the countryside a 30-year-old man noticed that his eyelids became red and swollen, lacrimation and nasal secre-

tions appeared, the voice became hoarse. He had a feeling of tightness in his chest. When he returned home, the intensity of these symptoms decreased, but they were still present.

1. Define the type of pathological process developed in the patient. Substantiate your opinion.
2. What methods can be used to determine the cause of this pathological process?
3. What are the main components of the mechanism of the given process?
4. What treatment approaches and preventive procedures can be used in this case?

Task 7.

A 38-year-old patient K. who had ingested gold-containing drugs according to the doctor's prescription noticed, first singular and then multiple, petechiae on the skin of his forearms, chest, back, and also in the oral cavity mucosa. Minor contusions were accompanied by extensive subcutaneous hemorrhages.

Blood tests showed normal count of erythrocytes, leukocytes, and Hb concentration but a significant decrease in platelets count, and an increase in the IgG and IgM content. Therapeutic procedures used by the doctor improved the patient's condition, and he continued to take gold-containing drugs since other forms of medication had poor effect.

1. Define the type of pathological reaction developed in the patient. Substantiate your answer.
2. What was the cause and mechanisms of this reaction? Substantiate your opinion by the given data.
3. What procedures could be used to inhibit such pathological reaction to the gold-containing drugs?

Task 8.

A 30-year-old female patient K. visited her dermatologist several times with complaints about the red itching spots on the face, neck and hands appear-

ing in the cold environment, such as windy cold weather. The physician prescribed her the ointment which alleviated these symptoms but did not eliminate them. One day, when she was in a hurry to get to her work, she washed her face with cold water. Thirty minutes after she had gone outdoors the sites of the skin exposed to cold water turned red, swelled and itching. The patient had to visit her doctor and later was admitted to hospital.

1. What pathological process(es) developed in the patient after she had washed her face with cold water? Substantiate your answer.
2. What is the mechanism of this process?
3. What pathological process should it be distinguished from and what is the key difference between these two processes?
4. What groups of medicines can be used to prevent or inhibit pathological process?

Task 9.

A patient with an open lower extremity trauma was repeatedly injected tetanus antitoxin in combination with antihistamines. On the ninth day after the last antiserum injection he presented with a high body temperature (up to 38°C), severe weakness, swelling and tenderness of the shoulder and knee joints; intensely pruritic disseminated urticarial eruptions, and the enlarged tender popliteal and inguinal lymph nodes.

1. What form(s) of pathology may be suspected in this patient?
2. What additional data are necessary for a final conclusion about the form of pathology which the patient suffers from?
3. Taking into account the given data, outline the possible cause and mechanisms of development of this pathology.
4. What approaches can be used to prevent the development of this pathological process?

Task 10.

On the sixth week of his stay in hospital after extensive myocardial infarction a successfully recovering patient started to suffer from a dull pain in the chest, which was aggravated by deep breathing movements, swallow, and

changes in the body position. He also presented with fever (the body temperature 39°C) and a pericardial friction rub. Blood tests showed the presence of eosinophilic leukocytosis and an increased titer of the «antimyocardial» antibodies. The physician made a diagnosis of postmyocardial infarction syndrome (the Dressler's syndrome).

1. Taking into account the immunogenic nature of the Dressler's syndrome, explain the type and origin of antigens causing this disease.
2. Define the type of a pathological reaction developed in the patient using the classification system of Gell and Coombs.
3. Describe the mechanism of this pathological reaction. To what type of immunoglobulin do the «antimyocardial» antibodies belong?
4. Prove (or disprove) the fact that the Dressler's syndrome represents the allergic reaction of the delayed type.

Task 11.

A patient with an extensive full-thickness burn of the thigh got i.v. infusion of blood plasma. Soon after this, he presented with hyperemia of the face and neck, psychomotor agitation, restlessness, fear of death, an intense throbbing headache, buzzing in the ears, and nausea. The physician suspected the development of allergic reaction and gave the patient an injection of antihistamine drug. However, the patient's state continued to worsen. He developed choking sensation, acute systemic hypotension (blood pressure 65/45 mm Hg), confusion, pallor, cold sweat, and finally lost his consciousness. Later, seizures and spontaneous urination occurred.

1. What pathological process (or reaction) and what type of it developed in the patient after the infusion of blood plasma? Substantiate your answer.
2. Name and characterize the main steps of pathogenesis of this pathological process.
3. Why did the parenteral injection of the antihistamine drug not improve the patient's condition? What procedures should be performed to prevent this pathological process (reaction)?
4. What factors caused the development of respiratory, hemodynamic

and psychoneurological disorders? Name these factors and characterize mechanisms of their action.

Task 12.

Patient M., male, 21 years old, suffers from hereditary form of hypogammaglobulinemia. His father also suffers from this disease; mother is healthy. An examination showed a significant decrease of IgM and IgG levels in the patient's blood.

1. What are the etiology, basic mechanisms and clinical manifestations of hypogammaglobulinemia?
2. Does the pattern of immune and allergic reactions change in hypogammaglobulinemia? Substantiate your opinion.

CLASS 6.

ALLERGY. ANAFILAXY

Learning objectives: study causes, mechanisms, and manifestations of various allergic reactions in the experiment and clinic.

Content:

1. Knowledge control and correction.

Questions and discussion:

1. Concept of allergy. Classification of allergic reactions. Gell-Coombs types of allergic reactions.
2. Concept of sensitization. Allergens, classification. Role of allergens in occupational diseases.
3. The general pathogenesis of allergic reactions.
4. Type I allergic reactions, examples, mechanisms. Mediators of allergy: types, role in the development of type I allergic reactions.
5. Type II allergic reactions, etiology, mechanisms, and examples. Mediators of type II allergic reactions.
6. Type III allergic reactions, etiology, pathogenesis, examples.
7. Type IV allergic reactions, etiology, mechanisms, and examples.
8. Allergies in childhood
9. Allergic processes in oral cavity, etiology, pathogenesis, principles of diagnosis and treatment.
10. Basic principles of prevention and treatment of allergic diseases. Types and mechanisms of hypo- and desensitization.

2. Assignments.

Experiment № 1. Anaphylactic shock in guinea pig.

Take preliminarily sensitized guinea pig that received 0.1 ml horse serum 14 days prior to the experiment. Inject a resolution dose (0,5-1,0 ml) of horse serum into the heart cavity or intravenously into the auricle or femoral vein. Observe the development of anaphylactic shock. Every 5 minutes note changes in respiration, measure rectal temperature and monitor anxiety, seizures, involuntary urination, and defecation. The results of the observations are recorded in

table.

In case of lethal outcome open a chest and observe the respiratory system and presence of the Auer-Lewis symptom. Record results in a protocol, answer questions and draw conclusions.

Table 3

**Results of the observations guinea pig at the development
of anaphylactic shock**

Indicators	Appearance	Mobility	Breathing rate	Rectal temperature	Involuntary reactions	Additional information
Time						
Initial state						
The after the administration of a resolving dose of serum						
After 5'						
After 10'						
Etc.						

1. What type of allergic reaction is anaphylactic shock according to the Gell-Coombs classification?

2. What stage of allergy did you observe? Explain the mechanism of the disorder.

3. What is the cause of death in case of lethal outcome? Explain the mechanism of "shock lung" development.

When answering questions, use the diagrams from the application (fig. 1, 2, 3, 4).

Experiment № 2. Indirect degranulation of mast cells.

Sensitize a white rat with 0.5 ml horse serum every other day for 3 days. Anesthetize the rat to the third stage of etheric anaesthesia. Place the rat on a dissection table in supine position; inject intraperitoneally 10 ml of 37 ° C saline. Massage abdominal wall for 1-2 minutes with fingers to obtain a peritoneal flush.

Open abdominal cavity with scissors along the abdomen midline with incision length about 3-3.5 cm. Take a chemical cup and rinse it with heparin so-

lution (1: 1000 dilution). Place rat's belly over the cup so that the peritoneal flush may flow down in the cup. Collect fluid and close abdominal wall layer wise.

Centrifuge the flush and gently remove supernatant.

Take three slides stained with neutral red and apply one drop of pellet to each with an eyedropper.

Add 1 drop of sensitized serum and allergen (horse serum) to the 1st slide and mix it gently; add 1 drop of sensitized serum and saline (serum control) to the 2nd slide; add 1 drop of allergen and saline (control of the allergen) to the 3rd slide. Lubricate corners of cover slips with vaseline and cover drops with cover slips. Incubate slides in the thermostat for 15 minutes.

Microscopy of samples/

Use lens x8 and eyepiece № 7 for microscopy of both controls. Find lilac colour mast cells among unstained leucocytes. Choose a field of view with a relatively large content of mast cells. Change the lens x8 to the lens x40. Find intact and degranulated mast cells in the established field of view.

Characteristics of intact cells. Diameter is 1.5 times larger than the diameter of leucocytes, regular round/oval shape, cytoplasmic lilac-red granules, and light nucleus. Membrane is continuous, clearly contoured when the microscope screw rotates.

Characteristics of degranulated cells. Uneven contour, vacuolated cytoplasm, stoma on the outer contour of the cell through which the granules pour out of a cell.

Calculate percentage of degranulated and intact mast cells in each drop. To do this, count cells while moving a slide along polyline until reach 50 mast cells. Note how many of cells are degranulated. Compare the obtained data.

Record results in a protocol and make conclusions based on the questions below:

1. What are the mechanisms behind specific degranulation of mast cells? What did cause it?

2. Which primary mediators (pre-existing and newly synthesized) are released during degranulation of mast cells, what role do they play in the devel-

opment of allergy? What is the role of secondary allergy mediators?

3. Is it possible for degranulation to develop in mast cells of non-sensitized animals? If so, in which cases, how do they differ from anaphylaxis?

Experiment № 3. Anaphylactic reaction of blood vessels and frog heart.

For this experiment you need frogs that were sensitized 25-30 days before the main experiment with a triple injection of 0.3 ml horse serum into the lymphatic sac. Frogs are kept at room temperature of 20-22 ° C.

A. Mesenteric anaphylactic reaction. Immobilize a sensitized frog by pithing; fix it on a board in prone position with its right side by the round hole. Make a flank incision and gently put out mesentery. Stretch it gently over the round hole of a plate. Note the initial circulation parameters including blood filling, lumen of blood vessels, and rate of blood flow. Select a site with good capillary circulation for observation because the reaction does not occur in large vessels. Apply 5 drops of a specific antigen (horse serum) to the mesentery and monitor rate of vascular reaction, extravasation of blood cells to the surrounding tissues, capillary stasis development.

1. Explain the mechanism of mesenteric vascular changes after exposure to specific antigen. Which allergy mediators have a key role in vascular reaction?

2. Define stage of the observed allergic reactions; describe possible consequences for the organism.

B. Anaphylactic reaction of the frog heart.

Fix immobilized sensitized frog in supine position to a board. Grip and raise sternum with tweezers, remove lower half of sternum, remove pericardium and expose heart. Grab the top of heart with a surfer connected to lever and record contraction of heart with a kymograph. Apply 3-6 drops of a specific antigen (horse serum) to heart and observe rhythm and amplitude changes until it stops. Then wash the heart with saline and observe restoration of its rhythm.

In 15 minutes apply 6 drops of the previous antigen to heart again and note reaction. Write down results and make conclusions.

1. Explain why there is a change in heart functioning of the sensitized an-

imal in this experiment?

2. What are the mechanisms of acute circulatory insufficiency in patients with systemic anaphylaxis?

When answering questions, use the diagrams from the application (fig. 1, 2, 3, 4).

TASKS

Task 1.

After spending an hour and a half in the countryside a 30-year-old man noticed that his eyelids became red and swollen, lacrimation and nasal secretions appeared, the voice became hoarse. He had a feeling of tightness in his chest. When he returned home, the intensity of these symptoms decreased but they were still present.

1. Define the type of pathological process developed in the patient. Substantiate your opinion.

2. What methods can be used to determine the cause of this pathological process?

3. What are the main components of the mechanism of the given process?

4. What treatment approaches and preventive procedures can be used in?

Task 2.

A 33-year-old patient L. who had ingested gold-containing drugs according to the doctor's prescription noticed, first singular and then multiple, petechiae on the skin of his forearms, chest, back, and also in the oral cavity mucosa. Minor contusions were accompanied by extensive subcutaneous hemorrhages.

Blood tests showed normal count of erythrocytes, leukocytes, and Hb concentration, but a significant decrease in platelets count, and an increase in the IgG and IgM content. Therapeutic procedures used by the doctor improved the patient's condition, and he continued to take gold-containing drugs since other forms of medication had poor effect.

1. Define the type of pathological reaction developed in the patient.

2. What was the cause and mechanisms of this reaction?

3. What procedures could be used to inhibit such pathological reaction to the gold-containing drugs?

Task 3.

A 30-year-old female patient S. visited her dermatologist several times with complaints about the red itching spots on the face, neck and hands appearing in the cold environment, such as windy cold weather. The physician prescribed her the ointment which alleviated these symptoms but did not eliminate them. One day, when she was in a hurry to get to her work, she washed her face with cold water. Thirty minutes after she had gone outdoors the sites of the skin exposed to cold water turned red, swelled and itching. The patient had to visit her doctor and later was admitted to hospital.

1. What pathological process(es) developed in the patient after she had washed her face with cold water?

2. What is the mechanism of this process?

3. What pathological process should it be distinguished from and what is the key difference between these two processes?

4. What groups of medicines can be used to prevent or inhibit this pathological process?

Task 4.

A patient with an open lower extremity trauma was repeatedly injected tetanus antitoxin in combination with antihistamines. On the ninth day after the last antiserum injection he presented with a high body temperature (up to 38 °C), severe weakness, swelling and tenderness of the shoulder and knee joints; intensely pruritic disseminated urticarial eruptions, and the enlarged tender popliteal and inguinal lymph nodes.

1. What form(s) of pathology may be suspected in this patient?

2. What additional data are necessary for a final conclusion about the form of pathology which the patient suffers from?

3. Taking into account the given data, outline the possible cause and mechanisms of development of this pathology.

4. What approaches can be used to prevent the development of this pathological process?

Task 5.

A patient with an extensive full-thickness burn of the thigh got i.v. infusion of blood plasma. Soon after this, he presented with hyperemia of the face and neck, psychomotor agitation, restlessness, fear of death, an intense throbbing headache, buzzing in the ears, and nausea. The physician suspected the development of allergic reaction and gave the patient an injection of antihistamine drug. However, the patient's state continued to worsen. He developed choking sensation, acute systemic hypotension (blood pressure 65/45 mm Hg), confusion, pallor, cold sweat, and finally lost his consciousness. Later, seizures and spontaneous urination occurred.

1. What pathological process (or reaction) and what type of it developed in the patient after the infusion of blood plasma? Substantiate your answer.

2. Name and characterize the main steps of pathogenesis of this pathological process.

3. Why did the parenteral injection of the antihistamine drug not improve the patient's condition? What procedures should be performed to prevent this pathological process (reaction)?

4. What factors caused the development of respiratory, hemodynamic and psychoneurological disorders? Name these factors and characterize mechanisms of their action.

5. What emergency treatment should be employed to help the patient with this pathological process (reaction)?

Task 6.

A female, 39 years of age, works as a secretary. For 6 months she has been working with a new copy machine. Over the last three months she has developed a blistering rash on both hands and the rash is now turned into deep scaling of the skin. There is a mild scaling at the site of both her earrings. The patient has used a bland cream with-out effect. The dermatologist examines her with patch tests, where solutions of common allergens, chemicals, metals including nickel are placed on her back and occluded with dressings. The next day (24 hours later) the dressings are removed and the exposed areas examined. The examination is repeated 48 hours after the beginning of the expo-

sure. There is a severe rash at the site of the colour powder used in the copy machine, and a mild rash at the nickel spot.

1. What is the diagnosis? Which therapeutic strategy is recommendable?

Task 7.

In a 5-year-old child suffering from diphtheria, 9 days after the administration of an antitoxic serocream, temperature rose to 39 ° C, erythematous rash, pruritus, and joint pain appeared. The level of complement in the serum is sharply reduced.

1. What pathology would you think about in this case?
2. What is the evidence of a decrease in complement activity in the serum of the patient?

Task 8.

Patient S., 25 years old, during a repeated course of vitamin therapy 10 minutes after intramuscular injection of vitamin B1 felt headache, dizziness, shortness of breath, abdominal pain. BP decreased to 60/40 mm Hg, pulse 120 in min, filiform.

1. What pathology would you think about? Is vitamin B1 an allergen?
2. Specify the type of reaction (according to Gell and Coombs) developed in the patient after injection of the drug.
3. What are the possible mechanisms of development of these symptoms/

Task 9.

In patient K., 25 years old, with traumatic damage to the left eye, after 4 weeks, the vision of the right eye deteriorated, sympathetic ophthalmia was diagnosed. Lymphocytes isolated from peripheral blood have the ability to inhibit migration of macrophages and stimulate the reaction of blast transformation.

1. What is the mechanism of development of pathology in this case? Give an informed explanation.

Task 10.

Patient M., 22 years old, came to the doctor with complaints about the burning sensation in the area of lips. There is periorbital erythema, insignificant infiltration, single small vesicles and small areas of wetness. Symptoms arose after 2 weeks of using a new imported lipstick.

1. What disease would you suspect?
2. How could you confirm the diagnosis without application of complex laboratory studies?
3. What is the mechanism of this pathological process?
4. What mediators matter in this pathology?

Task 11.

In the summer, an hour and a half after the arrival in the suburban zone a patient has blushed and swollen eyelids for 30 years, lacrimation, runny nose, hoarse voice, perspiration in the throat, difficulty breathing. Upon returning home to the city, these symptoms are preserved, although their severity was somewhat less.

1. How do you identify a pathological condition that has developed in a patient? Substantiate your answer.
2. How can you identify the cause of this condition?
3. What are the main pathways in the pathogenesis of this condition?
4. What approaches to therapy and prevention would you propose to use in this case?

Task 12.

Patient P., 10 years old, received 3000 units of antitetanus serum according to Besredka with a preventive purpose for a foot injury. On the ninth day after the introduction of serum, the child had severe pain and swelling of the shoulder and knee joints, with a generalized eruptions. At the same time, sharp weakness, deafness of heart tones, low blood pressure were observed. The child was hospitalized.

1. What allergic reaction has developed in a child?
2. What type of hypersensitivity is it?
3. What antibodies are responsible for the development of this allergic reaction?

CLASS 7.

CONTROL QUESTIONS

1. The concept of geriatrics and gerontology. Aging of the body. Theories of aging.
2. Features of pathological processes in elderly. Methods to prevent premature aging.
3. Role of heredity in pathology, methods of its study. Similarities and differences of acquired, congenital, heredity and family forms of pathology. Their etiology.
4. Fundamental regularities of gene diseases. Examples of gene heredity diseases with disorder of synthesis of transport, structure and enzyme proteins. Types of transmission of heredity diseases (autosomal-dominant, autosomal-recessive, codominant, intermediate and mixed).
5. Chromosome diseases- polyploidy, aneuploidy, their manifestations and pathogenetical characteristics. Possible ways of prophylaxis and methods of therapy of hereditary diseases. Value of environmental protection.
6. Conception of “constitution of organism”. Constitution as a form of group reactivity. Constitutional types: Classification, differences, and mechanisms of formation.
7. Value of constitution in pathology. Markers of constitution. Diathesis, types, value in pathology (lymphatico-hypoplastic, exudative and gouty diathesis).
8. Conception of reactivity. Types and mechanisms of reactivity. Forms of individual reactivity and factors determining it.
9. Resistance of the organism – passive and active, primary and secondary. Specific and nonspecific factors of defense. Pathology of nonspecific factors of defense (complement and properdin systems). Correlation of reactivity and resistance.
10. Immunodeficient conditions, their classification. Primary (hereditary and congenital) immunodeficiencies. Classification, manifestations, and consequences.
11. Secondary (acquired) immunodeficiencies and immunodepressive

disorders, their causes, principles of therapy.

12. Acquired immunodeficiency syndrome (AIDS). Etiology, prevalence, pathogenesis, consequences.

13. Definition of allergy, its actuality. Allergens as industrial disutilities and factors breaking ecological equilibrium of industrial enterprises. Classification of allergic reactions (Cook, Ado, Coombs). Interrelations between allergy and immunity.

14. Allergic reactions of 1 type (reagin). Stages, mediators, mechanisms of their action. Clinical manifestations (anaphylactic shock, atopic reactions).

15. Allergic reactions of 2 type (cytotoxic). Stages, mediators, mechanisms of their action, clinical manifestations.

16. Allergic reactions of 3 type (immune complexes). Stages, mediators, mechanisms of their action, clinical manifestations (serum disease, Artus phenomenon).

17. Allergic reactions of 4 type (cell-mediated). Stages, mediators, mechanisms of their action.

18. Autoimmune diseases. Etiology, pathogenesis. Role of external and internal factors in pathogenesis of autoimmune diseases.

19. Tissue incompatibility, its mechanisms. Transplantation, types. Conception of transplantation antigens and immunologic tolerance. Reaction “transplantant versus host”. Value of delayed allergic reactions in problem of transplantation.

20. Shock, characteristics. Similarity and differences of different types of shock. Stages of shock, fundamental structural and functional disorders on different stages of shock. Traumatic shock, etiology, pathogenesis.

21. Etiology and pathogenesis of different types of shock, their similarities and differences (septic, hemotransfusion, cardiogenic). Fundamental principles of prophylaxis and therapy.

22. Shock reactions (collapse, syncope). Definition, classification, pathogenesis. Similarity and differences between shock and collapse.

23. Coma, definition, classification, stages; neurologic primary coma, principles of therapy.

TYPICAL PATHOLOGICAL PROCESSES

CLASS 8.

TISSUE AND ORGAN CIRCULATION DISORDERS, DISORDERS OF MICROCIRCULATION

Learning objectives: study causes, mechanisms and manifestations of peripheral circulation disorders in experiment and clinic.

Content:

1. Knowledge control and correction.

Questions and discussion:

1. Concept and types of typical pathological processes.
2. Microcirculation: Concept, elements of microcirculatory system. Concept and types of disorders of peripheral circulation.
3. Arterial hyperaemia: Definition, types, signs. Mechanisms and consequences of arterial hyperaemia.
4. Venous hyperaemia: Definition, causes, signs, microcirculatory features. Tissue functional and structural changes in venous hyperaemia, consequences.
5. Ischemia: Definition, signs, microcirculatory features. Ischemic changes in tissues and mechanisms for eliminating them. Compensation for decreased blood flow in ischemia.
6. Bleeding: Causes, mechanisms, outcomes.
7. Embolism: Causes, types, blood circulation changes, outcomes.
8. Thrombosis and its causes. Mechanisms of thrombus formation in arteries and veins. Outcomes of thrombosis.
9. Stasis, its types, mechanism, and consequences. Rheological blood disorder leading to stasis in microvessels (disorders of erythrocyte compliance, microcirculation blood flow, erythrocyte sludge, changes in erythrocytes concentration within blood). Sludge phenomenon: Concept, causes, mechanisms, manifestations, consequences.

10. Role of occupational factors in circulation pathology.

2. Assignments.

Experiment № 1. Arterial hyperaemia in a frog's tongue.

Immobilize a frog by spinal cord destruction, do not decapitate it, and place it on a board in prone position so that its lower jaw is at the edge of triangular orifice. Open jaws with tweezers, fix lower jaw to the board with pins at corners of maw. Carefully straighten the tongue over triangular orifice by grabbing papillae with tweezers. Incline pins obliquely with heads outward so they do not interfere with lens of microscope. Correctly straightened tongue has hexagonal shape. Ensure that the tongue is not excessively stretched as this may impair its blood supply. Further, observe circulation under a microscope at low magnification. Pay attention to vessels' size, number of functioning capillaries, and rate of circulation. Then, carefully apply a drop of turpentine-oil mixture to the tongue without changing frog position. Study circulation changes and make a conclusion about their character. Wash turpentine-oil mixture off the tongue with a large amount of saline.

Describe features of microcirculation, characteristic for arterial hyperaemia and explain mechanism of occurrence.

Experiment № 2. Venous hyperaemia in a frog's tongue.

Use the frog prepared in experiment № 2. Visualize two pairs of parallel large blood vessels at sides of base of a tongue, whereas lateral vessel is a vein and medial one is an artery. To separate a vein from an artery pull edge of the tongue with tweezers. The vein will move laterally making it easy to place a ligature. Stab a needle with suture between artery and vein, make a knot WITHOUT tightening. Repeat for the second side of the tongue.

First, observe initial circulation under a microscope at low magnification. Further, tighten a knot over one of the veins without changing frog/tongue/microscope position. Observe circulation after a vein ligation. Then, ligate the second vein and pay attention to character of circulation; compare with the original appearance. Whilst observing note characteristic signs of venous hyperaemia, e.g. slowing of blood flow, vasodilatation, jerky and pen-

dulum movement of blood, stasis. Remove both ligatures and monitor restoration of blood circulation.

Draw conclusions by answering the questions:

1. What is the mechanism of pendulum movement of blood in venous hyperaemia?
2. Why does tissue blood supply decrease in venous hyperaemia, despite expansion of capillaries and increased intravascular pressure?
3. When and why venous hyperaemia can have a positive impact on a body?

Experiment № 3. Impact of acute blood loss on blood supply in a frog's tongue.

Examine initial blood circulation in tongue' vessels under a microscope. Further, amputate one of frog legs in upper third of a thigh. Observe blood supply changes in the tongue due to acute blood loss. Then, placed a ligature above the site of injury and inject 2 ml of 0.65% saline into subcutaneous lymphatic sac. Observe restoration of blood circulation under a microscope.

Make a conclusion about the mechanism of observed changes based on the following questions:

1. What stages of acute blood loss did you observe in the experiment?
2. What does cause spasm of peripheral blood vessels during blood loss? Is it a factor of damage or protection?
3. What was the purpose of saline administration?

Experiment № 4. Embolism of vessels in a frog's tongue.

In case of preserved blood flow in tongue' vessels you can make an experiment on the same frog. Keep the tongue attached to a board, turn the frog over. Grab the skin with forceps above the sternum and cut a triangular flap. Then, dissect muscles and ribs on both sides of the sternum, cut clavicles, and dissect sternum at the level of shoulder girdle. Visualize heart, remove pericardium, and cover the heart with a thin layer of moist cotton wool. Wrap tips of forceps with cotton and fix the base of heart with forceps. Inject 0,1-0,2 ml of lycopodium seeds suspension in ventricle. Turn the frog over. Observe emboli

advancing in vessels under the microscope. Note peculiarities of circulatory disorders in the frog's tongue.

Make a conclusion about the mechanism of observed changes based on the following questions:

1. What kind of blood circulation disorders occurs in embolism, what are the mechanisms?
2. Describe complications may occur in embolism, particularly in embolism of pulmonary trunk.

Experiment № 5. White and red thrombi in mesentery vessels.

Fix immobilized frog on a plate in supine position as close as possible to circular orifice. Make a layer wise incision on the lateral surface of abdomen for skin, muscles, and peritoneum. Carefully extract a loop of small intestine avoiding injury and twists. Stretch mesentery over round hole and fix it with pins. Stab the pins through the intestine in an inclined position with pin heads outwards. Observe initial circulation in mesenteric vessels, pay attention to vessels' diameter and blood flow rate. Select a medium-sized vein where blood flow has slowed for further investigation. Place a small crystal of sodium chloride with tweezers/pin and microscopy control near the chosen vein and observe thrombus formation; draw obtained observations.

Explain the mechanism of blood clot formation and answer following questions:

1. What conditions are necessary for thrombus formation (Virchow's triad)?
2. What is specific for the first phase of thrombus formation? What kind of thrombus does form during within agglutination phase?
3. What is the second phase of thrombus formation?
4. What is the third phase of thrombus formation?

TASKS

Task 1.

A 68-year-old male patient D. presented with chronic hepatitis and liver cirrhosis. The doctor introduced a needle into the abdominal cavity to perform aspiration of ascitic fluid. By the fifteenth minute, after 5 l of fluid had been removed, the patient felt bad, complaining of weakness, dizziness, and nausea. But the procedure was continued. After 1,5 l more fluid had been evacuated the patient developed syncope and lost consciousness. Several minutes later, after the emergency treatment, the patient regained total consciousness but still complained about weakness, dizziness, and nausea.

1. What was the doctor's mistake during the performance of ascitic fluid aspiration?
2. What are the mechanisms of syncope after the removal of ascitic fluid?
3. What are the possible mechanisms of adaptation of the brain circulation in this case?
4. Why did the adaptive mechanisms turn out to be insufficient in this patient?

Task 2.

A 56-year-old male patient presented with complaints about fatigability and pains in the gastrocnemius muscles when walking. The symptoms were relieved by rest. This is called the symptom of «intermittent claudication». In addition, he complained about increased sensitivity to cold, numbness, pins- and needles, tingling sensation in his legs at rest. The patient had a long history of heavy smoking since his teens. His occupation required working outdoors even in cold seasons when he sometimes suffered from cold. The patient's examination showed that skin on both soles was pale and felt cool and dry; the nails crumbled; no pulse was felt on posterior tibial arteries on both legs. The preliminary diagnosis was thrombangitis obliterans.

1. What form of organ circulation disorders is observed in the presented

patient? Name its characteristic features.

2. What are the possible causes and mechanisms of this form of the circulation disorders?

3. What is the possible outcome of the circulation disorder in this case?

4. What are the likely mechanisms of the development of each symptoms seen in the patient?

Task 3.

A patient's left leg is cool, pale, pulseless and painful. His right leg is warm, purple, swollen and painful, but still has a pulse.

1. Which leg is more likely to have an arterial thrombus, and which one – a venous thrombus? Explain.

2. What are causes (your differential diagnoses) for each type of occlusion?

Task 4.

A 61-year-old female patient had been suffering from arterial hypertension for many years. During the last 2 years she began to feel cold intolerance in the legs, numbness and pain in the gastrocnemius muscles on walking and then at rest. These symptoms were more intense at night and caused sleep disturbances. Six months ago she developed a skin lesion on the front surface of her right shin; later it transformed into ulcer. The ulcer was painless but resistant to therapy. During the visit to the doctor the patient complained about dryness in the mouth, constant feeling thirsty, frequent urination. On examination: the skin of the right shin was dry, pale, and cold; no pulsation of the sole artery was detected. Repeated blood tests showed increased levels of cholesterol, fibrinogen; high platelet counts; glucose 190-210 mg/dl. The urine is positive for ketone bodies and glucose.

1. What forms of pathology can be observed in the patient?

2. Name the likely type of lesion affecting the patient's arteries? Name and characterize the main mechanisms of pathology underlying changes in the patient's arterial wall.

3. What factors caused the development of erosion and ulcer on the patient's shin?

4. Can you assume the presence of the microcirculatory disorders in the vascular system of the patient's right leg? What are the likely causes of these disorders?

Task 5.

A 44-year-old male patient K. suffering from atherosclerosis and coronary heart disease resistant to pharmacological treatment underwent an excision of the upper cervical sympathetic ganglion and a dissection of the sympathetic nerves innervating the heart. Deep thermography of the myocard performed during the surgery revealed temperature normalization in previously «cold» (compared to the adjacent tissue) region of the middle third of the left ventricle anterior wall. The surgery results in a reduction: frequency and severity of angina pectoris episodes.

1. What are the possible mechanisms of the ischemic heart attacks in the patient? Substantiate your opinion.

2. How can you explain the fact of severe ischemia in only the middle third of the left ventricle anterior wall, despite the systemic nature of atherosclerotic lesions involving coronary arteries?

3. What caused a decrease in temperature in the «cold» (compared to adjacent regions) zone of the patient's heart before the surgery?

4. Can we consider the sludge phenomenon as one of the possible mechanisms of a reduced temperature in the ischemic regions of the heart? If you think so, explain its possible causes and mechanisms in this case.

Task 6.

Patient N, 63 years old, with stenosis of the right atrioventricular aperture, has acrocyanosis and minor edema of extremities. Biomicroscopic study of nail bed microcirculation revealed expansion of venous microvessels and outflow of erythrocytes beyond the vascular wall.

1. Describe the disturbances of regional blood circulation in this patient.

2. Explain the pathogenesis of described clinical manifestations.

Task 7.

An ambulance came 10 min after the vehicle accident, and the doctor started the emergency aid to the accident victim with a blunt trauma of the chest and open fracture of the right leg. A tourniquet was applied to the injured leg to stop severe hemorrhage; then an analgesic (morphine) was injected subcutaneously into the injured leg to prevent shock. In addition, the heart stimulating drugs were injected. However, despite the morphine injection the patient kept groaning of pain and suffered from progressing shortness of breath. The patient's blood pressure was 60/35 mm Hg, pulse rate 126 beats/min. Auscultation of the lungs revealed frequent and depressed breathing on the left side; no sounds were detected on the right side. The consciousness became obtunded. Cyanosis of the skin and mucosal membranes was progressing. The doctor subcutaneously injected a drug stimulating the respiratory center to activate breathing. However, it had no effect, and the patient lost consciousness.

1. What pathological processes developed in the victim of the accident? Substantiate your answer by the reported data.
2. Which of the processes is the most pathogenic? Explain its etiology.
3. Describe the main stages of pathogenesis of the mentioned pathological process.
4. Why did injections of morphine and the analeptic drug fail to improve the patient's condition?

Task 8.

A 36-year-old female patient who had been suffering from diabetes mellitus for more than 13 years presented with visual losses manifested as blurring of vision, «sand in the eyes» when reading small prints. On examination: a significant decrease in vision acuity, narrowing of lateral visual fields in both eyes; irregular thickening of the retinal microvessels wall; capillary microaneurysms and mural thrombi in the retinal microvessels; focal serous exudates and hemorrhages in the retina, and its neovascularization.

The doctor informed the patient that her vision impairment was the result of the diabetic microangiopathy. The latter includes pathological alteration of the orbital microvessel wall. The patient received appropriate advice and treatment.

1. What types of microcirculatory disorders can you distinguish in the patient's retina?

2. What specific types of microcirculatory disorders can arise from: capillary microaneurisms; irregular microvessels wall thickening; mural microthrombi.

Explain their pathogenesis.

3. Which form (or forms) of microcirculation disorders may give rise to focal retinal edema?

Can the microcirculatory changes described produce vision impairment in the patient? Explain the relevant mechanisms.

Task 9.

64-year-old patient with history of chronic ischemic heart disease and pronounced sudden pain in the left leg due to atherosclerosis has paleness of her skin. Pulse on the back of the left foot is not palpable. The limb is cold to touch. After 2 hours, pallor gave way to a sharp cyanosis.

1. What is the cause of aforementioned symptoms?

2. Explain origin of pain and cyanosis.

Task 10.

46-years-old patient with significant ascites underwent paracentesis. After removing 5 liters of fluid, the patient's condition suddenly and sharply deteriorated with dizziness and fainting developed. Patient passed out because of insufficiency of brain blood supply as a result of blood redistribution.

1. What are the effects in blood supply of the abdominal cavity resulted in ascites in the patient?

2. Why did blood redistribution occur after paracentesis?

Task 11.

The caisson worker R., 30 years old, during the decompression period felt itching in various body parts and pulsating pain in the muscles of shins. Immediately after decompression, he felt severe headaches, cervical veins swelled in the right hypochondrium, the skin became cyanotic and with a sticky sweat, sharp dyspnea developed. Arterial pressure could not be determined. A few minutes later, death occurred with signs of increasing right heart failure. At autopsy large bubbles of gas were found in cavities of the right heart.

1. What are the cause and mechanism of hemodynamic disorders that caused this fatal outcome?

Task 12.

Patient S., 25 years old, has pain and numbness in fingers when exposed to cold. During attacks, there is objectively sharp blanching of the skin of fingers and hands, a decrease in local temperature, and decreased skin sensitivity.

1. Describe disturbances of regional blood circulation and microcirculation in the patient.
2. Explain the pathogenesis of described clinical manifestations.

CLASS 9.

INFLAMMATION

Learning objectives: perform a pathophysiological analysis of experimental data and clinical cases for inflammation-based processes.

Class Content:

1. Knowledge control and correction.

Questions for a colloquium.

1. Definition of inflammation, its etiology. Etiologic factors of inflammation associated with occupation. Principles of classification of inflammation. The main components of inflammatory process. General and local signs of inflammation.

2. Primary and secondary alteration in inflammation. Metabolic changes, pathochemical and physico-chemical changes in the focus of inflammation.

3. Mediators of inflammation, types and origin, role in the development of secondary alteration and in general dynamics of inflammatory process.

4. Leucocyte migration in inflammation, its mechanisms, chemotactic factors. Role of leucocyte-endothelium interaction. Molecules of cell adhesion.

5. Phagocytosis, types, stages, mechanisms. "Metabolic burst" in phagocytes. Mechanisms of endocytosis and intracellular digestion. Oxygen-dependant and anoxic bactericidal mechanisms in phagocytes.

6. Reaction of microcirculatory vessels in inflammation. Changes in blood flow dynamics, stages and mechanisms.

7. Exudation. Mechanisms of inflammatory edema. Types and composition of exudates.

8. Acute phase reaction (response), its protective role in acute infections.

9. Theories of inflammation. Biological significance of inflammation. Dialectic relationship of damaging and protective inflammatory reactions. Pharmacotherapeutic principles of inflammation.

10. Etiology and pathogenesis of chronic inflammation.
11. Inflammation in newborns; features of vascular reaction, leucocyte migration, phagocytosis.
12. Peculiarities of craniomaxillofacial inflammation.

2. Assignments.

Experiment № 1. Amylolytic activity of serum exudates.

Centrifuge exudate to form a pellet from cellular elements. Carefully take liquid supernatant and place it in a separate tube. Make 1:10 dilution of this supernatant with saline. You now need to make serial dilution for 7 tubes with the prepared solution (1:10; 1:20; 1:40; 1:80; 1: 160; 1: 320; 1: 640).

To do so, take 7 tubes, mark them, and add 1 ml of the diluted serum (1:10) to the tubes 1 and 2. Further, add 1 ml of saline to the tubes 2-7. Take tube 2, shake it, take 1 ml out and transfer it to tube 3. Take tube 3, shake it, take 1 ml out and transfer it to tube 4. Follow this algorithm down to the tube 7. Take out 1 ml of mix from tube 7. Take a new tube, mark it as '8' and add 1 ml of saline to it, this tube is a control. Now you have series dilution: 1:10; 1:20; 1:40; 1:80; 1: 160; 1: 320; 1: 640.

Add 2 ml 0,1% starch solution to tubes 1-8, mix and put tubes for 30 minutes in a thermostat at + 37°C. Remove the tubes from thermostat and add 2 drops of Lugol's solution to each tube. Maximal dilution (minimal concentration) at which a complete starch digestion occurred corresponds to amylolytic activity of exudate.

1. What is the amylolytic activity of serum exudate?
2. Which exudate types have the greatest amylolytic activity?

Experiment № 2. Proteolytic activity of serum exudates.

Take 8 tubes and make serial dilutions as in the previous experiment. Then 2 ml of casein solution (or another protein) is put it in all test tubes. Put rack with test tubes in a thermostat for 30 minutes at +37°C. Further, remove tubes from thermostat and add 2 drops protein-binding reagent to each tube. This reagent binds casein and forms cloudy ring or flakes. Solution is transpar-

ent in tubes with complete protein digestion. Maximal dilution (minimal concentration) at which a complete protein digestion occurred corresponds to proteolytic activity of exudate.

1. What does determine proteolytic activity of exudate?

Experiment № 3. Vascular reaction in mesenterial inflammation of a frog's intestine (Conheim's Experiment).

Immobilize a frog by pithing without decapitation and place it on a board in prone position with right side by the round orifice. Open abdominal cavity on the right side along the anterior axillary line. Carefully (do NOT touch mesentery) take out a loop of the small intestine with tweezers and gently spread it around the round orifice of a plate. Fix intestine with pins to a board, do NOT stab mesenteric edge. Extraction of the intestine and its fixation to a plate are accompanied by mechanical injury, drying, chemical and bacterial impact, which causes development of acute inflammatory reaction in peritoneum. Therefore, when mesentery is ready for a study, you can see developing inflammatory vascular response.

Observe mesentery with low magnification of a microscope. Pay attention to small arteries, arterioles, capillaries, and venules. They are concentrated by the intestinal wall, whereas large vessels are located at the root of mesentery. Choose vessels without pigmentation and thick walls, you need to find a vessel with clear central stream of cellular elements and peripheral plasma stream without cells. Pay attention to changes in lumen diameter, flow speed and the width of the plasma space. The initially observed acceleration of blood flow is followed by slowing down. You may see impetuous and pendulum movement of blood flow. Within this period, leucocytes appear in the plasma space, you can see them as silver spheres rolling along vessel walls. Later leucocytes begin to occupy marginal position and adhere to the inner vessel wall.

Then cover mesentery gently with a cover slip and observe it with eyepiece 7 and lens x40.

Leucocytes lose rounded shape after marginal stand and begin to trans-

migrate through vessel walls into tissues. Observe changes for 1 -2 hours, draw different stages of vascular reaction and briefly record them.

1. What are the main theses of Conheim's doctrine of inflammation? State the sequence of changes in blood circulation within inflammatory locus.

2. At what stage of vascular changes does exudation begin? What are the conditions required for this?

3. How and why does vascular permeability change in the inflammatory focus? Which mediators are involved in?

When answering questions, use the diagrams from the application (fig. 5, 6).

Experiment № 4. Inflammatory tissue and vascular changes in a frog's tongue.

Use a frog from the previous experiment in prone position. Spread a tongue over the triangular orifice. Find an area near the edge of tongue with a well-defined capillary system. Put a small crystal of caustic silver (lapis, Ag-NO₃) on mucous membrane for a few seconds. Remove the crystal and rinse the tongue with distilled water. Observe changes in central and marginal parts of the lesion. Pay attention to peculiarities of blood flow in adjacent vessels (stasis, thrombosis, slowing of blood flow) and periphery (arterial vessel dilatation, acceleration of blood flow).

Draw conclusions about the relationship of alterative and vascular changes.

TASKS

Task 1.

A 46-year-old patient B. suffers from gastric ulcer. He has been admitted as an emergency into the surgery department with a preliminary diagnosis of gastric perforation. On admission: acute pain in the epigastric region, rigid abdomen, tenderness, signs of peritoneal irritation; body temperature 38,5°C; the absence of bowel sounds; marked leukocytosis; an increased erythrocyte sedi-

mentation rate. Analysis of the aspirated fluid (400 ml of opalescent fluid was evacuated during the needle drainage of the abdomen) showed the presence of a large amount of leukocytes, protein 4%, various types of bacteria including anaerobic. A plain radiograph of the abdomen demonstrated the presence of air under the diaphragm. The patient was transported to the operation room.

1. Can you confirm or disprove the preliminary diagnosis of gastric perforation? Substantiate your conclusion.
2. Define the type of pathological process in the patient on admission.
3. What are the causes and mechanisms of each of the symptoms presented by the patient?
4. Define the type of the fluid evacuated from the patient's abdominal cavity. Describe the characteristic properties of this fluid and explain the mechanisms of its formation.

Task 2.

Two industrial workers visited a primary care physician with heat burn of the shin area. They received them during the boiler breakdown accident. They presented with a similar set of complaints about headache, pain and swelling of the skin in the injured area. On examination: patient A. had erythematous shins with marked edema; patient B. had, in addition, blisters filled with a yellowish transparent fluid. Both patients received treatment recommendations, and were accorded a sick-leave. However, both of them ignored the doctor's advice. Three days later the patient A.'s condition improved, and the patient B.'s condition worsened. The latter developed extensive edema and pain in the injured areas, his body temperature increased up to 38,9 °C; the area of the heat burn was covered with numerous blisters filled with pus. Bacteriological analysis of the purulent content revealed the presence of staphylococci.

1. What pathological process(es) developed in the given patients?
2. What blood test can confirm the presence of this process in the victims of the industrial accident?
3. How can you explain the different course of development of the same

pathological process in the patients?

4. What mechanisms underlie the clinical symptoms in patient B.?

5. Why did the noninfectious pathogenic factor (high temperature) cause the emergence of blisters with pus in patient B.?

Task 3.

A 25-year-old patient A. visited gastroenterologist with complaints about acid belching, discomfort in the epigastrium, sensation of fullness in the stomach after meal, and frequent diarrhea. Gastroduodenoscopy and the complete blood count test with serum protein analysis were performed to specify the diagnosis. The results of the gastroduodenoscopy: gastric mucosa is swelled with enlarged, tortuous, closely adjoining folds; sulci between the folds are deep. Overt signs of arterial and venous hyperemia of mucosa and focal erosions are observed. The volume of mucus is increased. In the blood: signs of anemia and leukocytosis; dysproteinemia due to an increase in globulins fraction.

1. What pathological process(es) can be distinguished in this patient?

2. Substantiate your opinion.

3. Which of the pathological processes is primary, and which of them are secondary?

4. What are the possible causes and mechanisms of symptoms observed in the patient?

Task 4.

With an aim to perform a pathophysiological analysis of factors determining the unfavorable course of posttraumatic inflammation physicians of the traumatology department studied medical histories of their patients. All patients were distributed into three groups depending on the type of the observed physiological dysfunction. The first group included patients who exhibited signs of liver damage, the second group consisted of patients with cardiovascular pathology (atherosclerosis, essential hypertension, etc.), and the third group comprised elderly patients with various cerebral disorders (stroke, concussion of the brain, cerebrovascular atherosclerosis).

1. What pathophysiological grounds has distribution of patient three groups? Substantiate your opinion.

2. What other criteria (parameters) could be used to classify groups for studies of various aspects of inflammation?

Task 5.

A 68-year-old male patient D. presented with chronic hepatitis and liver cirrhosis. The doctor introduced a needle into the abdominal cavity to perform aspiration of ascitic fluid. By the fifteenth minute, after 5 l of fluid had been removed, the patient felt bad, complaining of weakness, dizziness, and nausea. But the procedure was continued. After 1,5 l more fluid had been evacuated the patient developed syncope and lost consciousness. Several minutes later, after the emergency treatment, the patient regained total consciousness but still complained about weakness, dizziness, and nausea.

1. What was the doctor's mistake during the performance of ascitic fluid aspiration?

2. What are the mechanisms of syncope after the removal of ascitic fluid?

3. What are the possible mechanisms of adaptation of the brain circulation in this case?

4. Why did the adaptive mechanisms turn out to be insufficient in this patient?

Task 6.

A 50-year-old patient B. suffers from gastric ulcer. He has been admitted an emergency into the surgery department with a preliminary diagnosis of gastric perforation. On admission: acute pain in the epigastric region, rigid abdomen, tenderness, signs of peritoneal irritation; body temperature 38,5°C; the absence of bowel sounds; marked leukocytosis; an increased erythrocyte sedimentation rate. Analysis of the aspirated fluid (400 ml of opalescent fluid was evacuated during the needle drainage of the abdomen) showed the presence of a large amount of leukocytes, protein – 4%, various types of bacteria including anaerobic. A plain radiograph of the abdomen demonstrated the presence of air

under the diaphragm. The patient was transported to the operation room.

1. Define the type of pathological process in the patient on admission.
2. What are the causes and mechanisms of the symptoms presented by the patient?
3. Define the type of the fluid evacuated from the patient's abdominal cavity and explain the mechanisms of its formation.

Task 7.

Patient T., 26 years old, noted pain, redness, and swelling developed 3 months following a subcutaneous injection in the right shoulder, . On the 7th day, swelling and soreness increased; there was fluctuation at palpation. Leukocytes in the blood – $20 \times 10^9/l$. ESR – 20 mm/h. After incision was made in the region of fluctuations, 10 ml of a yellowish-greenish liquid was evacuated. Her laboratory study noted high density protein content 0,7 g/l, pH 5,39, microscopy revealed a predominance of neutrophilic leukocytes.

1. What kind of inflammation developed in this patient?
2. What are the signs of inflammation?
3. Which of them take place in this patient?

Task 8.

Patient B., 12 years old and patient K., 43 years old, had fluid accumulation in the abdominal cavity. Both patients underwent paracentesis (abdominal puncture).

Patient B. had light yellow colored fluid with specific gravity 1.029 and protein content 0,39 g/l. In the sediment there are numerous blood cells with neutrophils of degenerative forms predominately. Microbial flora is located intra- and extracellularly.

Patient K. had a transparent fluid of light yellow color with a specific gravity 1.014 and protein content 0,2 g/l. In sediment there is a small number of cells, mainly lymphocytes.

1. What is the nature of fluids found in patient B. and patient K.? Substantiate your opinion.

2. What is the mechanism of fluid accumulation in abdominal cavity in these patients?

Task 9.

Patient M., 36 years old, complains about pain during swallowing, weakness, sweating, and headache. Body temperature 38,8°C. The mucous throat is hyperemic, edematous, covered with grayish membranes. Leukocytes are $15 \times 10^9/l$, ESR – 20 mm/h, pulse 98 bpm, respirations – 26 per min.

1. What pathological process does the patient have?
2. What signs of inflammation did occur in the patient? Explain the mechanisms of their origin.

Task 10.

Scientists performed an experiment on three anesthetized rats: they injected subcutaneously 0,1 ml of turpentine (a strong inflammatory agent) into the hind leg of each animal. The first rat served as a control, whereas the second had also 0.1 ml of diphenhydramine administered to the same leg in advance, and the third had 0,1 ml of hydrocortisone injected to the same leg in advance. The increment of the leg volume was evaluated after 30 minutes, after 1 hour and after 2 hours.

Results: in the first rat, the growth of leg volume was 30%, in an hour – 40%, after 2 hours – 90% compared to the initial volume. In the second rat, the growth was 2%, 35%, and 86% respectively. In the third rat, 5%, 15%, and 40%, res the foot of the rat? Explain the results.

2. Explain the mechanism of action of diphenhydramine and hydrocortisone.

Task 11.

Natasha K., 6 years old, arrived to clinic with a diagnosis of "Infectious inflammation of parotid salivary glands (mumps)." The disease began with a general malaise and a gradual increase in body temperature, which reached 39°C. Fever lasted for 10 days. The fluctuations between morning and evening temperatures did not exceed 1°C. After 10 days, the temperature gradually be-

gan to decrease, which was accompanied by increased sweating.

1. What pathological process did develop in the patient?
2. What type of temperature curve do you see in the patient?
3. What are the types of temperature curves?
4. What is the degree of temperature increase?
5. What stages of this pathological process do you observe in the patient?
6. Explain the mechanism of weakness, drowsiness, and decreased appetite.
7. Is it appropriate to administer aspirin to lower temperature?

Task 12.

A 25-year-old HIV-positive female patient was admitted to hospital with the body temperature 38,9°C, cough with rales. She complained about pain in her right chest during breathing. Laboratory tests showed leukopenia due to a decrease of lymphocytes and monocytes; bronchial secretions contained large amounts of desquamating epithelium, leukocytes, various strains of bacteria; blood was positive for treponema antigens.

1. What were the possible sources of pyrogens in this case?
2. How can you explain the development of fever in the patient suffering from leukopenia?
3. Can fever in this case be associated with AIDS?

CLASS 10.

TYPICAL METABOLIC DISTURBANCES.

DISORDERS OF WATER-ELECTROLYTE BALANCE

Learning objectives: study causes and mechanisms of typical disorders water metabolism.

Study role of the nervous system in experimental pulmonary edema.

Content:

1. Knowledge control and correction.

Questions and discussion:

1. Disorders of energy and basal metabolism. The role of the nervous and endocrine systems in pathogenesis.

2. Pathogenesis of water-salt metabolism disorders. Positive and negative water balance. Experimental reproduction of dehydration and hyperhydria.

3. Edema. Pathogenetic factors of edema: mechanical, membranogenic, oncotic, osmotic. The pathogenesis of cardiac, renal, inflammatory, toxic, allergic, starvation edema.

4. Main types and causes of dehydration. Physiologic changes in dehydration, consequences. Features of water metabolism in children.

5. Electrolyte disorders (sodium, calcium, magnesium). Main causes and mechanisms of electrolyte homeostasis disorders.

2. Assignments.

Experiment № 1. Adrenaline-dependant pulmonary edema in mice.

Weight two mice. Write down initial respiratory rate, amplitude of respiratory movements and behaviour for the both animals. Inject intraperitoneally 1 ml 0.1% solution of epinephrine to the first mouse and 1 ml of saline to the second (control) mouse. Write down time of injection. Check and write down respiratory rate and amplitude of respiratory movements every 5-10 minutes. Observe colour change for eyes and mucous membranes. The first mouse

should perish in 30-40 minutes after injection, sacrifice the mouse if not.

Fix the sacrificed mouse to a board with pins, open its chest and remove lungs. Put lungs on a watch glass and examine them. Pay attention to volume, colour, and hemorrhages. After that, detach lungs from trachea, heart and soft tissues, weight on scales and calculate the pulmonary coefficient.

$$\text{Pulmonary coefficient} = \frac{\text{Weight of lungs in grams}}{\text{Mouse weight in kilograms}}$$

For a healthy mouse pulmonary coefficient is 4-6 conventional units.

Typically, there is pulmonary edema in an experimental mouse.

1. What are the changes indicating pulmonary edema?
2. What are the main pathogenetic factors causing pulmonary edema with adrenaline administration?

Experiment № 2. Impact of oncotic pressure changes on edema and dehydration.

Immobilize two frogs by spinal cord destruction and pin them on boards in supine position. Open chest widely and pericardium for both frogs and put a ligature under the main branch of aorta. Fill a cannula with saline and insert it into the aorta in peripheral direction (away from heart) then ligate it at the level of neck. Dissect anterior abdominal wall along the middle line down to the pubic symphysis and continue incision to the flanks. This incision helps fluid to drain out of the abdominal cavity. Weight frogs with a board and a cannula and write down the value. Further, connect one frog via a rubber tube to a burette with saline, the other one to a burette with horse serum. Make sure that fluid levels and perfusion rates are identical in both burettes to ensure equivalent experimental conditions. You may regulate perfusion rate by a screw clamp on a rubber tube. One hour after the start of perfusion, detached the frogs from the burettes and weight them. Note difference in weight before and after perfusion for both saline and serum.

1. What is the mechanism of edema and dehydration in the experimental

frogs?

Experiment № 3. Impact of tissue osmotic pressure on edema.

You need two frogs for this experiment. Make injections in the dorsal lymphatic sac: 5 ml 10% NaCl solution for the first frog, 5 ml saline for the second frog. Place frogs in separate glasses, weight them and write down the values. After, pour water into both glasses. Wait for 1-1,5 hours, pour water away from the glasses and weight frogs again.

1. How would you explain weight change in a frog after introduction of hypertonic solution?

TASKS

Task 1

Examination of a 32-year-old patient revealed various signs of pathology, including excessive body mass: his height is 168 cm and weight 84.5 kg. The patient also has a pasty face, periorbital puffiness, pale skin; he had slow rebound of tissue to its original contour after pressing the feet or shin with the fingertip. The patient told the physician about tightness of a ring and shoes in the evening. The investigation of the cardiovascular system revealed the following: minor arterial hypotension, areas of cardiac dullness are slightly increased; other parameters are unremarkable. The daily urine volume is within the normal range.

1. What is the possible cause of the patient's excess of body mass?
2. Can we assume that water-ion balance is deranged in this patient?
3. What type of edema is observed in the patient?
4. What additional data are required to specify the type of edema in this case?

Task 2.

A 42-year-old patient has been admitted to hospital with a diagnosis of uncompensated chronic heart failure due to valvular disease. The patient has normal constitution with paucity of subcutaneous tissue. His height is 165 cm, body weight 81 kg. On examination: the patient needs to sit in bed; he has dyspnea, acrocyanosis, marked lower extremities edema, rales and wheezes

during auscultation of the chest. An X-ray investigation of the abdominal area shows an accumulation of fluid: the liver is enlarged; stroke volume and cardiac output are decreased; hematocrit 38%; the daily urine volume is decreased. Biochemical tests reveal an increased plasma activity of renin and an increased sodium concentration.

1. Are there any signs of derangement of water balance in this patient?
2. What type of dyshydria is observed in this case?
3. Is there any association between the accumulation of fluid in the subcutaneous tissue, the abdomen, and the lungs?
4. Explain pathogenesis of increased blood levels of renin and Na^+ in this patient.
5. Explain pathogenesis of edema in this patient.
6. Explain the role of edema in deterioration of the patient's condition.
7. What therapeutic approaches can be used to treat edema in this case?

Task 3.

A 22-year-old patient who recovered from severe scarlet fever two weeks ago complains about headache, pain in the back, dyspnea, and palpitations. During the last week she has increased her body weight by 11,5 kg. On examination: her face is pale; she has periorbital puffiness and edema of the shins and feet; the boundaries of the heart dullness are increased: blood pressure is 180/100 mm Hg; the daily urine volume is reduced. Urine tests show the presence of erythrocytes and protein. An increased titer of antistreptolysin O antibodies is found in the blood.

1. Is there evidence of the kidney damage in this patient? What is the possible mechanism of this pathology?
2. What is the cause of hyperhydration in this case: a decrease in water excretion or an increase in water retention?
3. Explain the mechanisms of edema in this patient.

Task 4.

A 7-year-old boy developed a progressive swelling of the soft palate with a swallowing difficulty, and then asphyxia after he had drunk mango juice. The

mucosal membrane in the swelled area is hyperemic without tenderness; a moderate increase in eosinophils is seen in the blood. The patient's body temperature is normal. His senior sister suffers from attacks of bronchial asthma.

1. Is edema in this case the result of ordinary inflammation?
2. What is the cause of edema in this patient?
3. Explain the pathogenesis of the given pathology.
4. Does this type of edema lead to life-threatening condition?

Task 5.

Patient N., 48 years old, for 10 years, suffers from chronic glomerulonephritis. Hospitalized in connection with the deterioration of well-being. He shows complaints about weakness, lethargy, apathy, bad sleep and appetite. Objectively, the skin is pale, with traces of scratching. There is swelling of the face, eyelids. The border of the heart is enlarged to the left. The heart sounds are muffled, rhythmic, accent P-th tone on the aorta, systolic murmur at the top. Pulse is intense. Blood pressure is increased. In blood tests: anemia, neutrophilic leukocytosis, increased ESR, hyperazotemia, hypoalbuminemia. In the analysis of urine: polyuria, isohypostenuria, moderate proteinuria.

1. What kind of edema (by etiology) was formed in the patient?
2. What are the mechanisms of its development?

Task 6.

Patient L., 37 years old, suffers from systemic lupus erythematosus. During the last exacerbation of the disease, massive, rapidly growing edema on the face, trunk and extremities first appeared. Laboratory tests revealed pronounced proteinuria, hypoproteinemia, hyperlipidemia.

1. What is the name of syndrome that is formed in the patient?
2. What are the mechanisms of edema development?

Task 7.

Patient D., 42, a teacher of higher education, returned from the next long trip to one of the countries of Africa. For several weeks, there were a series of short fits of fever with chills and increased sweating. Patients has also headaches, muscle pains, nausea, vomiting, and photophobia. After a while, the phe-

nomena of lymphangitis and lymphadenitis of the lower extremities joined with swelling of the legs. On examination: significant swelling of the lower extremities and external genital organs. The skin of the legs is hyperemic, edematous, dense elastic bands under the skin can be palpated in the region of thighs, groin, and scrotum.

1. What is the probable cause and mechanism of edema in this case?

Task 8.

Patient A., 38 years old, while at dacha, was bitten by a sow. Immediately after the bite, patient felt pain, a few minutes later blister, erythema, and severe itching at the site of bite appeared, and after a while patient had hives, nausea, spastic contractions of the abdominal muscles, expiratory dyspnea, massive ocular and vocal cords.

1. What kind of edema (by etiology) was formed in the patient?
2. What are the mechanisms of its development?

Task 9.

Patient M., 67 years old, with long history of ischemic heart disease for suffered extensive myocardial infarction five years ago. He has been hospitalized with complaints about swelling, resting dyspnea, attacks of suffocation at night, hemoptysis. Patient had pronounced acrocyanosis, swelling and increased pulse of cervical veins, massive edema on feet, legs, and lower back. The skin on shins is dense, dry with serous fluid flows out of skin fissures. In the lower sections of lungs soundless, damp rales are heard at both sides. Percussion reveals – sharp expansion of all borders of the heart. The abdomen is enlarged, there is free fluid in the abdominal cavity. The liver is significantly enlarged (the lower edge at the navel level), somewhat painful on palpation. Auscultation reveals irregular heart sounds (sinus arrhythmia), weak first tone and systolic murmur.

1. What violation of the water balance was formed in the patient?
2. What are the mechanisms of its development?

Task 10.

Patient K., 52 years old, was hospitalized with complaints about pain in

the right hypochondrium, skin itching, nosebleeds, jaundice. Skin is icteric with signs of scratching, hemorrhages, and spider angiomas. Abdomen is enlarged with caput medusae. Percussion – reveals signs of free fluid in the abdominal cavity. The liver span is reduced, tissue is dense. Spleen is enlarged and dense. Blood test reveals anemia, leukopenia, thrombocytopenia, hypoproteinemia, dysproteinemia and impaired functional liver tests. Patient has a history of viral hepatitis in severe form.

1. What kind of edema (by etiology) was formed in the patient?
2. What are the mechanisms of its development?

Task 11.

Patient T., 43 years old, was hospitalized with complaints about weakness, fatigue, weight loss, poor appetite, dyspnea with exertion, edema of legs. Examination reveals pronounced edema of face and neck ("Stokes collar") with cervical veins swollen. Heart sounds are significantly muffled, the early S3 tone (precordial stroke) is heard. Abdomen is enlarged, there are signs of free fluid in the abdominal cavity. Liver and spleen are enlarged. Patient has a history of pulmonary tuberculosis for several years. Additional studies confirmed the diagnosis of constrictive pericarditis of tuberculous etiology.

Explain the mechanism of edema of face and neck, peripheral edema and ascites in the patient.

Task 12.

One of the most constant and common signs in patients with nephritis is proteinuria, during the day the protein is lost up to 36 mg, while diuresis increases to 2000 ml. However, despite the excessive elimination of fluid edema develops in patients.

Explain the mechanisms of edema formation.

CLASS 11.

TYPICAL METABOLIC DISTURBANCES. DISORDERS OF PROTEIN, LIPID, CARBOHYDRATE METABOLISM

Learning objectives: study causes and mechanisms of typical disorders in carbohydrate, fat, protein metabolism.

Content:

1. Knowledge control and correction.

Questions and discussion:

1. Disorders of energy and basal metabolism. The role of the nervous and endocrine systems in pathogenesis.

2. Disorders of the main stages in protein metabolism. Positive and negative nitrogen balance.

3. Types and stages of fasting. Metabolic and functional changes in starvation. Peculiarities and consequences of protein and caloric deficiency in childhood. Impact of congenital and acquired defects of the oral cavity in development of incomplete starvation.

4. Amino acids exchange disturbances: Causes, role in pathology. Gout: Etiology and pathogenesis.

5. Disorders of carbohydrate metabolism, hypo- and hyperglycaemia, causes and types. Experimental models of insulin insufficiency.

6. Diabetes mellitus, its types. Disorders of carbohydrate and other metabolic and physiological functions in diabetes mellitus. Diabetic coma (ketoacidotic, hyperosmolar), pathogenetic features.

7. Disorders of the main stages in fat metabolism, fatty infiltration and dystrophy. Hyperketonemia. Types of obesity. Features of lipid metabolism disorders in children.

8. Disorders of cholesterol metabolism, hypercholesterolemia. Role of lipid metabolism disorders in the atherosclerosis.

9. Vitamin disorders. Features, etiology, pathogenesis of rickets and hypervitaminosis D in children. D-resistant forms of rickets.

TASKS

DISORDERS OF CARBOHYDRATE METABOLISM

Task 1.

A 50-year-old patient F. was admitted to hospital in an unconscious state. On examination: the skin appears dry, turgor of the skin and eye ball is reduced: shallow breathing; heart rate 96 beats/min; the tongue is dry; recurrent cramps of the extremities and face muscles. Blood tests show hyperglycemia (20mM glucose), hyperazotemia, hypernatremia, pH 7,32.

The relative who accompanied the patient to hospital told the physician that the patient had been suffering from diabetes mellitus and had ingested small doses of hypoglycemic medicines. During the last month he experienced exacerbation of chronic cholecystitis and colitis, suffered from vomiting and diarrhea which occurred quite often. The patient also felt constantly thirsty and frequently urinated.

1. Define the pathological state of the patient on admission.
2. What was the cause of this pathological state? Describe the main stages of its pathogenesis.
3. Why do patients lose consciousness during the development of this pathological state and the similar ones?
4. What methods are used to treat patients in this pathological state?

Task 2.

A 23-year-old male patient visited his physician with complaints about intermittent muscle weakness, dizziness, headache, episodes of poor vision, tremor of hands, irritability, and, occasionally, confusion. The paroxysms of this condition occurred more frequently during the last four months. The patient attributed his malaise to the psychological stress that accompanied his professional activity, and also associated it with an acute feeling of hunger. After

evaluation of the patient the physician made a diagnosis of neurasthenia and recommended an appropriate treatment. However, the disease went on progressing, and 1,5 months later the patient was brought by ambulance to the emergency room with a diagnosis of coma of unclear etiology.

On admission: consciousness is lost; there is evidence of midriasis, muscle cramps; tachycardia, arterial hypotension; irregular breathing; blood glucose level 30 mM.

1. What form of pathology caused the clinical manifestations presented by the patient during his first visit to the physician?
2. Define the pathological state of the patient in the emergency room.
3. What are the main steps of pathogenesis of this pathological state?
4. What other forms of pathology should be kept in mind when you are making a differential diagnosis of this pathological state?

Task 3.

A 43-year-old patient C. with an excessive body mass suffers from diabetes mellitus. To control his blood glucose he uses hypoglycemic drugs. Two weeks before he was admitted to hospital he had had an episode of excessive alcohol drinking, and soon after that he noticed an enhanced feeling of dry mouth. He drank a lot of water (up to 8-10 l daily) and urinated frequently. He felt general weakness and pain in the legs. On the night before the admission the patient was delirious. When he woke up in the morning he was agitated, rest-less, and confused. The patient's relatives called in the ambulance. On admission: consciousness is absent; the skin is dry and pale. The results of the blood tests: glucose 45mM, lactic acid 2,9 mM, pH 7,29. Urine glucose level 2 mM.

1. What other signs, in addition to the observed in this patient, may be found in a) diabetes mellitus and b) diabetic coma? Describe these signs and explain their pathogenesis.
2. What are the main stages of pathogenesis of this coma?

Task 4.

A female patient, approximately 40 years of age, had been brought to the emergency room in an unconscious state. Witnesses of the incident told the physician that she had suddenly lost her consciousness in the bus on her way back from her summer house in the countryside. A card that was found among her papers indicated that she suffered from diabetes mellitus and took a slow-release form of insulin.

On examination: consciousness is absent, corneal and deep ten-don reflexes are not observed, the pattern of breathing is unremarkable; blood pressure 80/60 mm Hg; tachycardia is present; the skin appears moist: turgor of the eye balls is increased; general trembling is alternating with episodes of clonic and tonic seizures.

The patient was treated with insulin, but no improvement was observed: the patient's condition even worsened: breathing became irregular, blood pressure decreased to 70/50 mm Hg, tachycardia progressed, and the duration of seizures increased.

1. What pathological state developed in the patient before and after administration of insulin?
2. Describe causes and mechanisms of this state. What therapeutic procedure would have been appropriate for patient before and after the administration of insulin?

Task 5.

In two monozygous sibs (8 months of age) an examination revealed a significant increase in the liver size (hepatomegaly), decreased fasting level of blood glucose (hypoglycemia), and lack of fasting blood glucose changes after the adrenaline administration. The liver biopsy investigation showed an increased content of glycogen and a significantly diminished activity of glycogen phosphorylase in hepatocytes.

1. Give the definition of the pathological process developed in the infants.

2. What are the possible causes of this process?
3. Explain the sequence of events leading to glycogen accumulation in the liver cells in this form of pathology.
4. Explain the association, if any, between the low activity of glycogen phosphorylase in the patients' liver and a) an increased content of glycogen in the hepatocytes; b) hepatomegaly; c) a decreased fasting blood glucose levels; d) lack of hyperglycemic effect of adrenaline.

Task 6.

A 61-year-old female patient had been suffering from arterial hypertension many years. During the last 2 years she began to feel cold intolerance in his legs, numbness and pain in the gastrocnemius muscles on walking and then at rest. These symptoms were more intense at night and caused sleep disturbances. Six months ago she developed a skin lesion on the front surface of her right shin; later it transformed into ulcer. The ulcer was painless but resistant to therapy. During the visit to the doctor the patient complained about dryness in the mouth, constant feeling thirsty, frequent urination. On examination: the skin of the right shin was dry, pale, and cold; no pulsation of the sole artery was detected. Repeated blood tests showed increased levels of cholesterol, fibrinogen; high platelet counts; glucose 20-25 mM. The urine is positive for ketone bodies and glucose.

1. What forms of pathology can be observed in the patient?
2. Name the likely type of lesion affecting the patient's arteries? Name and characterize the main mechanisms of pathology underlying changes in the patient's arterial wall.
3. What factors caused the development of erosion and ulcer on the patient's shin?
4. Can you assume the presence of the microcirculatory disorders in the vascular system of the patient's right leg? What are the likely causes of these disorders?

Task 7.

A 34-year-old female patient who had been suffering from diabetes mellitus for more than 13 years presented with visual losses manifested as blurring of vision, «sand in the eyes» when reading small prints. Dr. examination: a significant decrease in vision acuity, narrowing of several visual fields in both eyes; irregular thickening of the retinal – microvessels wall; capillary microaneurisms and mural thrombi in the retinal microvessels; focal serous exudates and hemorrhages in the retina, and its neovascularization.

The doctor informed the patient that her vision impairment was the result of the diabetic microangiopathy. The latter includes pathological alteration of the orbital microvessel wall. The patient received appropriate advice and treatment.

1. What types of microcirculatory disorders can you distinguish in the patient's retina?

2. Which form of microcirculation disorders may give rise to focal retinal edema? Can the microcirculatory changes described produce vision impairment in the patient?

Task 8.

A 19 year old male, body weight 80 kg, was suddenly complaining of fever during his work and ordered home to bed. The patient was living alone. Fortunately a colleague visited him the next morning. He had to break the door down and found the patient unconscious. The patient arrived at the hospital in deep coma. The urine contained glucose and ketone bodies. Explain the condition of the patient concerning thermo-balance, carbohydrate metabolism.

Task 9.

A 23-year old male was saved after 30 days in the ruins of a house following earth quake. There was no food but sufficient water. At the arrival to the hospital the patient was in syncope with frequent, deep respiration, and the expired air smelled of acetone. The skin was dirty with brown pigmentation. The cardiac rate was 85 bpm, and the arterial blood pressure was 85/55 mmHg.

The blood glucose was 2,2 mM, and the plasma FFA was increased. The

serum concentrations of proteins and essential amino acids were reduced. The blood hemoglobin was 95 g/l. There was moderate antidiuresis with ketonuria with signs of water retention and a high nitrogen loss in the urine.

The patient was treated with parenteral administration of glucose, amino acids and electrolytes. Following the glucose intake, the blood glucose was increased to 10 mM, and glucosuria occurred. A glucose tolerance test was performed and resulted in a high blood glucose level that had not reached the normal level within 2 hours.

1. Describe the energetic events leading to survival.
2. Why did the patient smell of acetone?
3. What happened to the carbohydrate metabolism of the patient?
4. Explain the high nitrogen loss in the urine.

Task 10.

Patient K., 10 years old, was referred to an endocrinologist with complaints about intermittent headaches, a feeling of thirst that has been bothering him for the last 2 months. The results of the examination in the hospital: 160 cm height, lean, a liver and a spleen are enlarged, heart chambers are enlarged. Blood pressure is 140/90 mm Hg. The total content of proteins and fatty acids is increased, the level of glucose in the blood is 14 mmol/l, glucose tolerance is lowered. Daily diuresis – 4 liters.

1. What disorders of endocrine system can be assumed?
2. What is the mechanism of hyperglycemia and its possible consequences?
3. Describe and explain changes in metabolism of the patient.

Task 11.

Three patients were admitted to the hospital for examination.

Patient A had fasting hyperglycemia, normal glucose tolerance, increased insulin in the blood, polyuria, glucosuria.

Patient B had low fasting glucose, normal glucose tolerance, high insulin, polyuria, glucosuria.

Patient C had normal fasting glucose, insulin, and glucose tolerance. Af-

ter taking 200 g of glucose, hyperglycemia and glucosuria were noted.

1. Determine types of hyperglycemia and hypoglycemia in each patient.
2. Explain the mechanism of hyperglycemia, polyuria, glucosuria.

Task 12.

After an intensive training, the athlete has sharply worsened condition. He has general weakness, mental excitement, trembling extremities, pallor, and intense diaphoresis. He complained about acute hunger, feeling of heartbeat, numbness of the lips, double vision (diplopia).

1. What changes in metabolism did cause this condition?
2. Explain the pathogenesis of symptoms.
3. What are the possible consequences of this condition?
4. Give practical recommendations for management of this condition in setting of emergent care.

DISORDERS OF LIPID METABOLISM.

DISORDERS OF PROTEIN METABOLISM

Task 1.

During his visit to the doctor a 55-year-old male patient complains of anginal pain which recently has increased in frequency and intensity. The patient is the president of a big trading company. During the last months he has experienced great psychological stress, spent 12-14 hours at his office daily and slept little; he began smoking more than one pack of cigarettes per day. The patient is a hearty eater, preferring high-calorie fat food. His father died of myocardial infarction. On examination: his height is 173 cm, weight 89 kg, blood pressure 175/100 mm Hg. Blood tests show the following: total plasma cholesterol – 6,8 mM, triglycerides – 1,9 mM. HDL cholesterol 0.8 mM, LDL cholesterol – 5,3 mM; glucose tolerance is decreased.

1. What form of pathology is the cause of angina pectoris in this patient?
2. What risk factors of this pathology are observed in the patient?
3. Does the patient have signs of lipid metabolism disorder? If he does,

what forms of it?

4. What therapeutic approaches can be used to treat lipid metabolism disorders in this case?

Task 2.

A 13-year-old boy visited a doctor with complaints about recurrent pain in tire region of the heart. The pain was exacerbated at exertion. Examination of the patient showed the presence of small, firm, yellow nodules (xanthomas) over the course of the hand muscles tendons, the presence of corneal lipid arcus. Angiographic examination revealed the coronary artery stenosis. Blood tests results demonstrate: total plasma cholesterol – 22 mM, triglycerides – 1,7 mM, HDL-cholesterol – 0,7 mM (normal values: >0,9 mM), LDL-cholesterol – 9 mM (normal values: 3,0-4,5 mM). The patient's parents also have increased plasma levels of cholesterol – more than 8mM. Immunocytochemical analysis of the patient's leukocytes revealed abnormality of the LDL-receptor.

1. What type of hyperlipidemia is observed in the patient?

2. Is heredity important in the development of this pathology? If it is, what is its type of inheritance? What is the prevalence of this pathology in general population?

3. What mechanisms underlie hyperlipidemia in this patient?

4. Assess the atherogenicity of the patient's plasma by calculating the index of atherogenicity.

5. What are the pathological consequences of hyperlipidemia of this type? What therapeutic approaches may be used to treat this pathology?

Task 3.

During examination of a 5-year-old boy a physician found an enlargement of the liver and spleen (hepatosplenomegaly), swelling of the tonsils which had red-orange color, and an enlargement of inguinal, axillary, and other superficial lymph nodes. Also clouding of the cornea was observed.

Biochemical tests revealed: total cholesterol of the blood plasma 0,1 mM, triglycerides – 2,3 mM, HDL fraction almost absent. During electropho-

retic separation of the patient's plasma apoproteins A-I and A-II bands were not observed.

1. What type of disorders of lipid metabolism is observed in the patient?
2. What mechanisms underlie the development of this disorder?
3. Why is HDL fraction considered antiatherogenic? What factors determine HDL levels in the blood plasma?
4. What role do apoproteins of the A class play in the cholesterol metabolism?

Task 4.

During experimental studies of atherosclerosis one group of rabbits was fed cholesterol added to a standard pelleted food in a dose of 5 g per day. Cholesterol was previously purified to remove contaminating oxides. The second group was treated similarly, but cholesterol was not purified, and the third group received the same diet with cholesterol that was previously subjected to peroxidation by hydrogen peroxide in the presence of ferrous chloride (FeCl_2). Six months after the start of the experiment the rabbits were killed by a lethal dose of a narcotic drug; the abdominal portion of the aorta and coronary arteries were isolated and their internal surface were examined for the presence of lesions. Simultaneously blood samples were obtained from all rabbits of the experimental groups; plasma was separated from formal particles and used to isolate LDL fraction. In the following in vitro studies samples of the isolated LDL fractions from various groups were added to a media containing cultures of murine peritoneal macrophages, and the extent of intracellular accumulation of cholesterol esters was assessed following incubation for three days. LDL fraction of the intact rabbits was used as a control.

1. Which group of the rabbits is supposed to incur the maximum damage to their arteries due to the excess of cholesterol in the food?
2. Describe the pathways by which LDL can enter peripheral cells, including macrophages.
3. What differences in accumulation of lipids in murine macrophages can

be expected with LDL fraction isolated from various experimental animals?
Compare LDLs of the control and experimental groups of rabbits.

4. What role do monocytes/macrophages play in atherogenesis?

Task 5.

Patients K., 45 years and M., 45 years old, were diagnosed with dyslipoproteinemia (DLP): in patient K. blood levels of total cholesterol, LDL cholesterol, and HDL cholesterol were increased; in the patient M., the content of total cholesterol, LDL cholesterol, and the content of HDL cholesterol were increased in the blood.

1. At whom from subjects above the probability of development of an atherosclerotic process (explain, why?)

2. What role does LDL and HDL and the characteristic apoproteids play in the transport of cholesterol?

3. How is the cholesterol coefficient determined? What does it show?

Task 6.

Two groups of patients with coronary artery disease and high cholesterol participate in clinical trial the first group takes cholestiramin; the second takes placebo for 5 years. Coronarography was performed before and after the study,. In patients taking cholestyramine, coronary artery stenosis was less common, and the number of myocardial infarctions was less by 19% compared to the control group.

1. What does determine the pronounced atherogenic effect of modified LPs?

2. Specify the sequence of atherosclerotic plaque formation.

3. Explain the mechanism of action of cholestyramine and positive results in patients from the experimental group.

Task 7.

Pregnant patient N., 25 years, due to toxicosis developed acute dystrophia of the liver with liver failure. The patient is confused, delirious, there are muscle twitchings. Skin and mucous membranes are icteric. Pulse 92/min, blood pressure 100/70 mm Hg. Blood and urine amino acids and ammonia are

sharply increased, urea is decreased. Special studies have revealed blood of biogenic amines (histamine, tyramine, serotonin) in large amount.

1. What data does indicate failure of protein metabolism?
2. What are the possible causes and mechanisms of these disorders?
3. Does the patient have a disruption of the final stage of protein metabolism?
4. Justify your conclusion. What causes the patient to be excited?

Task 8.

The patient S., 46 years old, complains about paroxysmal pain in joints of fingers and toes. Pain is accompanied with fever. Examination reveals hypersthenic physique. There are small round nodules (tofuses) on cartilages of auricles. Foot joints are sharply deformed. Xrays of metatarsophalangeal joints demonstrate well-defined non-structured circular foci articular surfaces. Blood urate is 1,9 mmol/l.

1. What kind of disease would you think about this case?
2. Explain pathogenesis of these symptoms.
3. Accumulation of which metabolic products led to the development of arthritis?

Task 9.

Patient S., 11 years old, arrived to a district pediatrician with complaints about fatigue, weakness, and sore throat. The girl had numerous infectious diseases: sore throat, rhinitis, respiratory infections, pneumonia. The patient was diagnosed with catarrhal angina. Total serum protein was 72,0 g/l, albumin 71%, α 1-globulin 8,5%, α 2-globulin 8,2%, β -globulin 7,3%, γ -globulin – 5%. Blood lymphocytes are 16%.

1. How could you estimate parameters of protein spectrum in blood serum?
2. Why does the patient often have frequent infectious diseases?
3. Which tests can be done to prove the diagnosis?

Task 10.

Female patient adheres to strict diet in order to achieve the desired standard of body shape. She refuses to eat and induces vomiting. For last 4

monthspatient lost 20 kg, had aggressiveness caries, and dysmenorrheal, and was diagnosed psychogenic anorexia. Examination revealed height 178 cm, weight 45 kg, poor development of subcutaneous fat layer, pale skin and visible mucous membranes.

Blood test: Hb 100 g/l, erythrocytes $3.2 \times 10^{12}/l$, color index 0.8, leukocytes $3.3 \times 10^9/l$, ESR 3 mm/h. Blood biochemistry : total protein 59,0 g/l, A/G 0,96 (albumins 49%, globulins 51%), fibrinogen 4,4 g/l, cholesterol 6,7 mmol/l, hyperaminoacidemia.

1. What types of metabolic disorders could be identified in this patient?
2. What are the possible causes and mechanisms of these disorders?
3. What stages of protein metabolism are failed in the patient?
4. Explain the pathogenesis of these symptoms and laboratory changes.
5. Assess the degree of body weight deficit and the form of starvation.

Task 11.

Patient P., 25 years old, complained about general weakness, weight loss (5 kg for 6 months), frequent loose abundant stools, pain in umbilical area, flatulence. 10 months ago she got salmonellosis and was treated in the infectious department. Soon after discharge, she began to notice a penchant for diarrhea. She treated herself with levomycetin. Examination revealed weight 52 kg, height 165 cm, white-coated tongue, smoothened papillae. Umbilical area is painful to palpation and swollen.

Blood test: erythrocytes $3.8 \times 10^{12}/l$, hemoglobin 110 g/l, leukocytes $8.1 \times 10^9/l$, ESR 18 mm/h. Total protein 65 g/l. Stool test: pulpy consistency, yellow color , offensive odor, undigested muscle fibers (++), neutral fat (+), soaps and fatty acids (++), grains of extracellular starch (++), a lot of mucus.

1. What typical disorders of protein metabolism are present in the patient?
2. What are the possible causes and mechanisms of these disorders?
3. What stages of protein metabolism are failed in the patient?
4. Name syndromes explaining these symptoms and laboratory changes

with.

5. What additional biochemical studies do you need for reassurance?
6. Substantiate your opinion.

Task 12.

Patient R., 49 years old, admitted to oncology clinic with gastric cancer. He complains about general weakness, weight loss, lack of appetite, vomiting, diarrhea, and edema. Examination reveals exhausted condition, pale skin, facial edema, swollen feet. There is lack of free hydrochloric acid in gastric secret and large amount of lactate. Plasma protein is 40 g/l.

1. Name this symptom complex.
2. Explain the pathogenesis of these clinical and laboratory symptoms.
3. Explain change in plasma protein and name its cause.
4. Specify the vicious circle of pathogenesis in this case.

CLASS 12.

TYPICAL METABOLIC DISTURBANCES.

DISORDERS OF ACID-BASE BALANCE

Learning objectives: to study disorders of acid-base balance, mechanisms of acidosis and alkalosis, and their complications.

Content:

1. Knowledge control and correction.

Questions and discussion:

1. Mechanisms of acid-base balance maintaining. Role of buffer systems, lungs and kidney in the acid-base balance regulation.
2. Parameters of acid-base balance and their importance. Sig-gart-Andersen nomograms.
3. Classification of acid-base disturbances. Compensatory and decompensatory, absolute and relative, respiratory and metabolic disorders.
4. Causes and parameters of acid-base balance in: a. respiratory acidosis; b. metabolic acidosis; c. respiratory alkalosis; d. metabolic alkalosis.
5. Consequences of acid-base disturbances. Clinical manifestations of alkalosis and acidosis.

Table 4

Normal parameters of the acid-base homeostasis of the capillary blood

Parameter	Value
pH	7,36-7,45
pCO ₂	6-45 mm Hg
B	21-25 mmol/L
BB	45-52 mmol/L
BE	+/- 2,4 mmol/L
Ketone bodies	0,5-2,5 mg/dl
Lactic acid	6-16 mg/dl
Titrateable acidity (of the daily urine)	20-40 mmol/d
NH ₄ (of the daily urine)	30-50 mmol/d

TASKS

When answering questions, use the diagram and tables from the application.

Task 1.

The patient was on the high altitude (3000 m) about 3 days. After arterial blood gas analysis the parameters of acid-base balance were following:

$\text{pH} = 7,47$ $\text{pCO}_2 = 33,4$ mm Hg

$\text{HCO}_3^- = 20$ mM

$\text{BB} = 41,4$ mM

$\text{BE} = 2,0$ mM

Make a conclusion about the type of acid-base balance disorder and about the cause of this disturbance.

Task 2.

The patient with peptic ulcer was treated by sodium bicarbonate. After arterial blood gas analysis the parameters of acid-base balance were following:

$\text{pH} = 7,48$ $\text{pCO}_2 = 53,4$ mm Hg

$\text{HCO}_3^- = 31$ mM

$\text{BB} = 55$ mM

$\text{BE} = 5$ mM

Make a conclusion about type of disorder of acid-base balance and about the cause of this disturbance.

Task 3.

The patient with diabetes mellitus presents in reanimation in coma state. Arterial blood gas analysis reveals changing in the acid-base parameters. Make a conclusion about the type of this disorder.

$\text{pH} = 7,28$ $\text{pCO}_2 = 31,4$ mm Hg

$\text{HCO}_3^- = 21$ mM

$\text{BB} = 36,4$ mM

$\text{BE} = -8,5$ mM

Make a conclusion about the type of acid-base disorder.

Task 4.

The patient with bronchial asthma was treated in pulmonary department of hospital. The arterial blood gas analysis revealed the following changes in the acid-base parameters:

$$pH = 7,29 \quad pCO_2 = 51,4 \text{ mm Hg}$$

$$HCO_3^- = 31 \text{ mM}$$

$$BB = 44,4 \text{ mM}$$

$$BE = -2,5 \text{ mM}$$

Make a conclusion about the type of acid-base disorder.

Task 5.

The patient has had the hysteria hyperventilation episode just before the analysis. $pH = 7,57 \quad pCO_2 = 28 \text{ mm Hg}$

$$SB = 24 \text{ mM}$$

$$BE = +2,5 \text{ mM}$$

$$BB = 48 \text{ mM}$$

Titrateable acidity of urine – 20 mmol/d

Make a conclusion about the type of acid-base disorder.

Task 6.

The patient suffers from an acute left ventricular insufficiency with lungs edema. $pH = 7,17 \quad pCO_2 = 51 \text{ mm Hg}$

$$SB = 24 \text{ mM}$$

$$BB = 45 \text{ mM}$$

$$BE = 2,5 \text{ mM}$$

Make a conclusion about the type of acid-base disorder.

Task 7.

The patient has hepatic coma. Make a conclusion about the type of acid-base disorder.

$$pH = 7,11 \quad pCO_2 = 24 \text{ mm Hg}$$

$$SB = 15,5 \text{ mM}$$

$$BE = -13 \text{ mol/L}$$

BB = 38 mM

Ketone bodies of the blood – 33 mM

Titratable acidity urine – 70 mmol/d

Task 8.

The patient suffers from diabetes mellitus. Make a conclusion about the type of acid-base disorder.

pH = 7,36 p pCO₂ = 31 mm Hg

SB = 19.5 mM

BB = 39 mM

BE = -5 mM

Ketone bodies in blood –10 mM

Titratable acidity in urine – 37 mmol/d (N=10-30 mmol/d)

Task 9.

The patient is undergoing surgical operation with cardiopulmonary by-pass. Make a conclusion about the type of acid-base disorder.

pH –7.20 pCO₂ – 32 mm Hg

SB – 14 mM

BB – 29 mM

BE – -12 mM

Task 10.

The patient is admitted to hospital with a preliminary diagnosis of an acute myocardial infarction. Make a conclusion about the type of acid-base disorder.

pH –7.30 pCO₂ – 32 mm Hg

SB – 18 mM

BB – 36 mM

BE – -6 mM

Lactic acid in blood – 10 mM

Task 11.

The patient is undergoing surgery with mechanical ventilation. Make a

conclusion about the type of acid-base disorder.

pH – 7.52 pCO₂ – 25 mm Hg

SB – 21 mM

BE – -2 mM

BB – – 44 mM

Task 12.

The patient with diabetes mellitus in coma. Make a conclusion about the type of acid-base disorder.

pH – 7,09 pCO₂ – 51 mm Hg

SB – 15 mM

BE – -12.5 mM

BB – 38 mM

Lactic acid (blood) – 15 mM

Titrateable acidity of urine – 18 mmol/d

CLASS 13.

CONTROL QUESTIONS

1. Arterial hyperemia, etiology, signs, mechanisms of development. Peculiarities of microcirculation, outcomes and consequences of different types of arterial hyperemia.

2. Venous hyperemia, etiology, signs, mechanisms of development. Peculiarities of microcirculation. Consequences of venous hyperemia, pathophysiological reasons of use venous hyperemia in therapy.

3. Local anemia, stasis. Etiology and mechanisms of development, consequences for the organism. Meaning of collateral circulation.

4. Thrombosis. Thrombogenic factors and mechanisms of thrombus formation. Types of the thromboses. Outcomes of thromboses.

5. Embolism. Etiology and mechanisms of embolus formation. Types of embolism. Mechanisms of disorders originate from embolism.

6. Bleedings, types. Compensatory-adaptive and pathologic reactions at hemorrhage. Methods of emergency hemorrhage stop and first aid.

7. Definition of inflammation, etiology. Basic components of inflammation process. General and local signs of inflammation.

8. Primary and secondary alteration at inflammation. Peculiarities of changes in metabolism, pathochemical and physicochemical changes in locus of inflammation.

9. Mediators of inflammation. Types and their origin. Role of mediators in development of secondary alteration and in general dynamics of inflammation process.

10. Emigration of leukocytes at inflammation, mechanisms, factors of chemotaxis. Phagocytosis, types, stages and mechanisms. Pathology of phagocytosis.

11. Reaction of microcirculatory vessels at inflammation. Dynamics in changes of bloodstream, stages and mechanisms.

12. Exudation. Mechanisms of genesis inflammatory edema. Composition and types of the exudates.

13. Theories of inflammation. Principles of classification inflammation. Biological role of inflammation. Interaction between damage and compensatory-adaptive reactions in inflammation process.

14. Fever, definition, etiology. Pyrogenic substances, their structure and origin of formation. Differences between fever, exogenous overheating and another types of hyperthermia.

15. Stages of fever. Thermoregulation on different stages of fever. Changes in metabolism and physiological functions at fever. Types of fever and types of temperature curves at fever. Biological value of fever reaction.

16. Neoplasms, role of chemical, physical and viral blastomogenic factors. Biological peculiarities of benign and malignant neoplasms. Distinction between normal and neoplasm cell.

17. Common theories of neoplasm development. Contemporary conception about molecular mechanisms of cancerogenesis. Value of oncogenes, role of oncoproteins in cancerogenesis.

18. Antineoplasm resistance of organism – anticancer, antimutation and anticellular mechanisms. Paraneoplastic syndrome as an example of interaction between neoplasm and host. Principles of prophylaxis and therapy of neoplasms. Mechanisms of resistance neoplasms to the therapeutical actions.

19. Types and stages of starvation. Changes of metabolism and organism functions at starvation.

20. Disturbances of energy and basal metabolism. Role of nervous and endocrine systems in pathogenesis of this disorders.

21. Disorders of basic stages of protein metabolism. Positive and negative nitrogen balance.

22. Etiology of disorders of amino acid metabolism, their role in pathology. Etiology and pathogenesis of gout.

23. Disorders in carbohydrate metabolism, hypo- and hyperglykemia,

etiology and types. Experimental models of insulin insufficiency.

24. Diabetes mellitus, its types. Disorders of carbohydrate and another types of metabolism, and physiological functions at diabetes mellitus.

25. Disorders of basic stages of fat metabolism. Fatty infiltration and dystrophy. Hyperketonemia. Types of obesity.

26. Disorders of cholesterol metabolism. Hypercholesterinemia. Role of disorders of fatty metabolism in development of atherosclerosis.

27. Pathogenesis of disorders in water-salt metabolism. Positive and negative water balance. Experimental procreation of dehydration and hyperhydration.

28. Edema. Pathogenetic factors of edema: mechanical, membranogenic, oncotic, osmotic. Pathogenesis of cardiac, kidney, inflammatory, toxic, allergic and hungry edemas.

29. Basic types and etiology of exsiccation. Changes in the organism in condition of exsiccation, consequences.

30. Disorders of electrolyte metabolism (sodium, calcium, magnesium). Basic reasons and mechanisms of disorders in electrolyte homeostasis.

31. Disorders of acid-base equilibrium. Acidosis, alkalosis and their types. Disturbances in the organism under different types of acidosis and alkalosis. Compensatory reactions of the organism at the disorders of acid-base equilibrium.

32. Liposoluble vitamins and pathologic processes linked with their excess and insufficiency.

33. Water soluble vitamins and pathologic processes linked with their excess and insufficiency.

TESTS

As a result of training, students master General professional (GPC 9) and professional competences (PC 1, 6, 21)

Choose one correct answer

1. WHAT DOSE OF ALLERGEN CAN CAUSE ANAPHYLACTIC SHOCK IN SENSITIZED GUINEA PIGS

1. Primary
2. Resolving
3. Sensitizing
4. Lethal

2. INDICATE WHICH OF THE FOLLOWING DISEASES THAT CAN DEVELOP IN A TYPE IV IMMUNE INJURY:

1. Hashimoto's Thyroiditis
2. Tuberculin skin test
3. Acute glomerulonephritis
4. Autoimmune hemolytic anemia

3. MARK, WHAT ORGANS AND TISSUES CAN'T BE ATTRIBUTED TO THE "BARRIER ORGANS":

1. The tissue of the lens of the eye
2. The fabric of testicles
3. The colloid of the thyroid gland
4. The red blood cells

4. WHAT CELLS FORM THE BASIS OF THE INFLAMMATORY INFILTRATE IN ALLERGIC REACTIONS, DEVELOPING ON I THE TYPE OF IMMUNE DAMAGE:

1. Neutrophils
2. Lymphocytes
3. Eosinophils
4. Monocytes

5. POINT OUT THAT A MAJOR PATHOGENIC LINK HYPOGLYCEMIC COMA:

1. Carbohydrate "starvation" attack
2. Carbohydrate and energy starvation of brain neurons
3. Hypoosmia of blood
4. Uncompensated ketoacidosis

6. IN THE PATHOGENESIS OF AUTOIMMUNE DISEASES, AN IMPORTANT ROLE IS PLAYED BY ANTIBODIES BELONGING TO THE CLASS OF IMMUNOGLOBULINS:

1. IgE
2. IgG₁
3. Ig A
4. Ig G₄

7. SELECT WHICH CLASS OF IMMUNOGLOBULINS ARE ANTIBODIES IN ALLERGIC REACTIONS, DEVELOPING II AND TYPE III IMMUNE INJURY:

1. IgE
2. IgG₄
3. Ig A
4. Ig M

8. SPECIFY WHICH RESPONSE AND HUMAN DISEASE IS NOT ATTRIBUTED TO ATOPIC

1. Pollinosis
2. Urticaria
3. Angioedema
4. Serum sickness

9. ALLERGIC REACTIONS, MAINLY DEVELOPING UNDER THE SECOND TYPE OF IMMUNE DAMAGE, ARE:

1. Serum sickness
2. Angioedema
3. Autoimmune hemolytic anemia
4. Systemic lupus erythematosus

10. INDICATE ALLERGIC REACTIONS, EMERGING TYPE IV IMMUNE INJURY:

1. Contact dermatitis
2. Food Allergy
3. Serum sickness
4. Hashimoto's Thyroiditis

11. SPECIFY THE OPTIMUM TIME TO PLAY ANAPHYLACTIC SHOCK OF GUINEA PIGS AFTER ACTIVE SENSITIZATION:

1. 15-20 minutes
2. 24 to 48 hours
3. 6-8 days
4. 14-15 days

12. THE STATE OF SENSITIZATION IS CHARACTERIZED BY:

1. The Rash on the skin
2. Local tissue edema
3. The Increase in the titer of specific antibodies or the number of T-lymphocytes
4. A Disorder of the circulatory system and breathing

13. THE MECHANISM OF REDUCTION OF FEVER:

1. Increase of heat production
2. Increase heat transfer
3. Balance between heat generation and heat transfer
4. Increasing heat transfer

14. SPECIFY THE REACTION, NOT DEVELOPING FOR I-TYPE (REAGINIC) IMMUNE DAMAGE:

1. Urticaria
2. "Dusty" bronchial asthma
3. Serum sickness
4. Angioedema

15. INDICATE WHICH OF THE FOLLOWING DISEASES CAN DEVELOP IN TYPE II IMMUNE INJURY:

1. Hashimoto's Thyroiditis
2. Tuberculin reaction
3. Contact dermatitis
4. Autoimmune hemolytic anemia

16. SPECIFY THE DISEASE, AN ESSENTIAL LINK OF THE PATHOGENESIS OF WHICH ARE AUTOIMMUNE REACTIONS:

1. Atopic form of bronchial asthma
2. Immune form of agranulocytosis
3. Post-Traumatic "sympathetic" inflammation of the eyeball
4. Pollinosis

17. SPECIFY DISEASES NOT RELATED TO AUTOIMMUNE:

1. Hashimoto's Thyroiditis
2. Arthus
3. Rheumatoid arthritis
4. Systemic lupus erythematosus

18. INDICATE WHICH OF THE FOLLOWING DISEASES THAT CAN DEVELOP IN A TYPE IV IMMUNE INJURY:

1. Hashimoto's Thyroiditis
2. Autoimmune hemolytic anemia
3. Contact dermatitis
4. Acute glomerulonephritis

19. WHAT ARE THE REASONS OF THE VENOUS HYPEREMIA?

1. Compression of the arteries
2. Compression of the veins without damaging arteries
3. Violation of the integrity of vasoconstrictor nerves
4. Stasis

20. WHICH BIOLOGICALLY ACTIVE SUBSTANCES CAN CAUSE AN ARTERIAL HYPEREMIA?

1. Acetylcholine

2. Catecholamine
3. Bradykinin
4. Thromboxane A₂

21. ENDOGENOUS PYROGENS

1. Interleukin-1
2. Immunoglobulins
3. Antigen+antibody complex
4. Acute phase proteins of inflammation

22. IDENTIFYING THE FACTORS CAUSING HYPERGLYCEMIA:

1. Prevalence of braking processes in the Central nervous system
2. Digestive enzyme deficiency
3. Insufficiency of the pancreas
4. Limiting the intake of carbohydrates with food

23. IDENTIFYING THE FACTORS THAT CAUSE GLUCOSURIA IN DIABETES:

1. Ketonuria
2. Ketonemia
3. Reduction of the renal threshold for glucose excretion
4. Microangiopathy of renal glomeruli capillaries

24. INCREASING OF ORGAN AND TISSUE'S TEMPERATURE IN THE ARTERIAL HYPEREMIA AREA IS CAUSED BY:

1. An increased inflow of arterial blood
2. Attenuation of oxidative processes
3. Intensification of lymph formation (chylopoiesis)
4. An increased number of functioning capillaries

25. AN ARTERIOVENOUS OXYGEN DIFFERENCE IN THE BLOOD IN VENOUS HYPEREMIA IS:

1. Increasing
2. Decreasing
3. Not changing
4. Initially increases, then decreases

26. WHAT DOES NOT APPLY TO SIGNS OF ISCHEMIA

1. Decreasing of temperature
2. Decreasing of organ volume
3. Pulsation of the small blood vessels
4. Blanching

27. SPECIFY THE ORGAN WHERE THERE ARE NO FUNCTIONALLY ABSOLUTELY INSUFFICIENT CAPILLARIES

1. Brain
2. Skeletal muscles
3. Heart
4. Spleen

28. ARTERIAL HYPEREMIA IS NOT CHARACTERIZED BY:

1. Increasing of number of functioning capillaries
2. Decreasing of intracapillar circulation
3. Increasing of circulatory speed in capillaries
4. Increasing of lymph drainage from tissues

29. WHAT DOES NOT APPLY TO SIGNS OF VENOUS HYPEREMIA:

1. Increasing of organ volume
2. Cyanosis
3. Blanching
4. Cooling

30. WHAT CAN CAUSE AN ISCHEMIA?

1. Arterial obstruction by thrombus
2. Veins compression by scar tissue
3. Violation of the integrity of peripheral nerves
4. Weakening of lymph circulation

EXAMPLES OF TASK'S SOLUTION

Class 1. Subject and methods of pathophysiology Basic concepts of general nosology. Influence of environmental factors in the occurrence of diseases (radiant energy, overload, acceleration).

Task 5.

Physicist T., 35, was taken to the clinic 5 days ago with complaints about headache, dizziness, nausea, and vomiting. As a result of the accident, he was subjected to a general irradiation with gamma rays at a dose of 200 BER (2 Gr). At the moment he feels well, he does not make any complaints. CBC shows leukocytopenia, lymphocytopenia, decreased amount of reticulocytes.

1. What period of radiation sickness is present in the patient?
2. How would you explain patient's complaints at the admission?
3. What is the mechanism of changes in blood and what are the consequences of these changes?

Answers:

1. The patient has a second period of bone marrow form of acute radiation sickness—a period of imaginary well-being.

2. Upon admission, the victim complained of headache, dizziness, nausea, vomiting. These symptoms are the result of direct damaging effects of ionizing radiation on the epithelial cells of the digestive system and bone marrow, as well as the result of violations of neurohumoral regulation (there is increased excitability of the nervous system, lability of vegetative functions, activation of the pituitary-adrenal system).

3. Changes in the blood associated with direct damage to bone marrow cells, this leads to its devastation, disruption of the production of progenitor cells myeloid, lymphoid, erythroid Roskov, reducing the content of leukocytes, reticulocytes and erythrocytes in peripheral blood. With a lack of leukocytes and lymphocytes, the immune system will suffer, infectious processes develop, control over their mutated damaged cells is violated. Deficiency of reticulocytes and erythrocytes leads to hypoxia, oxygen starvation of tissues. Metabolic processes are disrupted, energy deficit occurs, cell structures are damaged. Suffer the functions of organs and tissues, the most sensitive brain and heart muscle.

Class 2. Causal relationships in pathogenesis Damaging effects of changes in barometric pressure. Oxygen Starvation.

Task 6.

An accident occurred with the aircraft during its flight at the height of 10,000 m. The cabin of the aircraft lost its hermetic state, and in several seconds the barometric pressure inside fell down to atmospheric level of 170 mm Hg at this altitude. The crew failed to make an urgent lowering maneuver, and the aircraft continued the flight at the same height for several minutes.

1. What pathological processes developed in the body of passengers during the aircraft accident? Which of them were the most dangerous for their well-being and can lead to death?

3. What are the likely causes and mechanisms of these pathological processes?

4. What types of preexisting pathology could include the highest risk for life in this situation?

Answers:

1. As a result of depressurization of the cabin at an altitude of 10,000 m, passengers experience acute general exogenous hypobaric hypoxia. In addition, the health and life of passengers can be threatened by a rapid drop in atmospheric pressure, cooling of the body, as well as stress due to the emergency situation.

2. The reason for the general exogenous hypoxia in this case-a sharp decrease in the partial pressure of oxygen in the inhaled air (as a result of reducing atmospheric pressure in the cabin depressurized). A condition that significantly aggravates the state of hypoxia is a rapid drop in barometric pressure (this factor in itself causes a special pathological condition, referred to as "decompression syndrome"). The main links in the pathogenesis of exogenous hypobaric hypoxia are hypoxemia, hypocapnia and the resulting additional deterioration of blood supply to the heart and brain, alkalosis, ion imbalance, and increased tissue oxygen demand. Later, it is possible to develop interdependent and other types of hypoxia: circulatory (due to a decrease in the contractile function of the heart and circulatory disorders), respiratory (as a result of pulmonary edema), tissue (due to disorders of biological oxidation), hemic (due to

changes in the affinity of hemoglobin to oxygen in the conditions of physical and chemical status of the blood).

The stress state is the result of the action on the body of a complex of extraordinary in nature and intensity factors: hypoxia, hypobaria, cold (at an altitude of 10,000 m the air temperature outside the aircraft is much lower than zero), as well as – conscious passengers factor direct threat to their lives due to the possibility of destruction and fall of the aircraft.

3. The greatest danger to life in this situation is the General acute severe hypoxia, the rapid development of which reduces the possibility of effective implementation of the compensatory mechanisms of the body. In addition, the danger to the life of aircraft passengers is the development of severe pathogenic stress, in which often there are circulatory disorders due to the occurrence of myocardial infarction (especially in the elderly) or ventricular fibrillation.

Class 6. Allergy. Anaphylaxis.

Task 1.

After spending an hour and a half in the countryside a 30-year-old man noticed that his eyelids became red and swollen, lacrimation and nasal secretions appeared, the voice became hoarse. He had a feeling of tightness in his chest. When he returned home, the intensity of these symptoms decreased but they were still present.

1. Define the type of pathological process developed in the patient. Substantiate your opinion.
2. What methods can be used to determine the cause of this pathological process?
3. What are the main components of the mechanism of the given process?
4. What treatment approaches and preventive procedures can be used in?

Answers:

1. Pathological condition, developed in the patient, referred to as hay fever (lat. pollen-pollen). This group of diseases is united on a causal basis – all of them are caused by pollen of plants (herbs, shrubs, some trees). In this case, an allergic reaction developed after arriving in a country recreation area, where there were obviously asthenia, pollen which caused allergic conjunctivitis, rhinitis, tracheobronchitis.

2. A specific substance (a component of pollen) can be detected by means of a skin test application of pollen of different plants to the scarified skin area. With a positive reaction in the scarification area there is a pronounced swelling, redness, bubbles may form.

3. Hay fever to develop according to the mechanism of reactions of type I (according to Gell a. Coombs).

This mechanism includes several stages. In the immune stage for the first time in the body allergen is captured and "processed" antigen-presenting cells (a-cells). Subsequently, as a result of the interaction of A-cells, T – and b-lymphocytes, the latter synthesize allergic antibodies (IgE and IgG4), which are fixed on the target cells of I order (mast cells, basophils and other leukocytes). At the stage of pathochemical reactions, target cells of the I order form and secrete biologically active substances-mediators of the allergic reaction. This determines the development of clinical manifestations (stage of clinical manifestations). This patient develops signs of conjunctivitis, rhinitis, tracheobronchitis.

4. The main principles of therapy and prevention of allergic reactions are: 1) etiotropic (allergen detection and prevention contact); 2) pathogenic (specific immunotherapy, antimediatory therapy, immunotherapy); 3) symptomatic (elimination of unpleasant and painful sensations).

The main principle of treatment is pathogenetic, consisting in specific immunotherapy (SIT: by repeated intradermal administration of small, gradually increasing doses of the allergen causing the damage reaction).

THE ANSWERS TO THE TESTS

1.	2	16.	2
2.	2	17.	2
3.	4	18.	3
4.	3	19.	2
5.	2	20.	3
6.	2	21.	1
7.	4	22.	3
8.	4	23.	3
9.	3	24.	1
10.	1	25.	1
11.	4	26.	3
12.	3	27.	2
13.	2	28.	2
14.	3	29.	3
15.	1	30.	1

LIST OF RECOMMENDED LITREATURE

Main literature:

1. Pathology / E. F. Goljan. Rapid Review 4th ed. – Elsevier, 2014. – 777 p.
2. Pathophysiology / Lee-Ellen Copstead, Jacquelyn Banasik. 5th ed. – Elsevier, 2013. – 1222 p.
3. Robbins & Cotran Pathologic Basis of Disease / V. Kumar, A. K. Abbas, Jon C. Aster. 9th ed. – Elsevier, 2014. – 1408 p.
4. Robbins & Cotran Review of Pathology / Edward C. Klatt, Vinay Kumar. 4th ed. – Elsevier, 2015. – 492 p.
5. Robbins Basis Pathology / V. Kumar, A. K. Abbas, Jon C. Aster. 10th ed. – Elsevier, 2018. – 910 p.
6. Understanding Pathophysiology / Sue E. Huether, Kathryn L. McCance, Valentina L. Brashers, Neal S. Rote. 6th ed. – Elsevier, 2017. – 1139 p.

Additional literature:

1. General and clinical pathophysiology / ed. by A.V. Kubyshkin. – Vinnytsa: Nova Knyha Publishers. – 2011. – 656 p.
<http://medsouls4you.blogspot.com>
2. Pathophysiology: Concise lectures, test, clinic-pathophysiological situations and clinic-laboratory cases. Students manual / P.F. Litvitsky, S.V. Pirozhkov, E.B. Tezikov. – Moscow «Geotar-Media», 2013. – 432 p.
3. Robbins and Cotran Pathologic Basis of Disease [Electronic resource]: application to the textbook on CD-ROM / V. Kumar, A. K. Abbas, N. Fausto. – Elsevier, 2005.
[http://library.bashgmu.ru/jirbis/index.php?option=com_irbis&Itemid=300&I21DBN=BSMU&I21DBNAM=BSMU&C21COM=S&S21ALL=\(%3C.%3E I=%E8%2F122-%D1%C4%3C.%3E\)](http://library.bashgmu.ru/jirbis/index.php?option=com_irbis&Itemid=300&I21DBN=BSMU&I21DBNAM=BSMU&C21COM=S&S21ALL=(%3C.%3E I=%E8%2F122-%D1%C4%3C.%3E))
4. Understanding Pathophysiology / K. L. McCance, Sue E. Huether. 5th ed. – Mosby, 2011. – 1184 p.

APPLICATION

Theme: Allergy

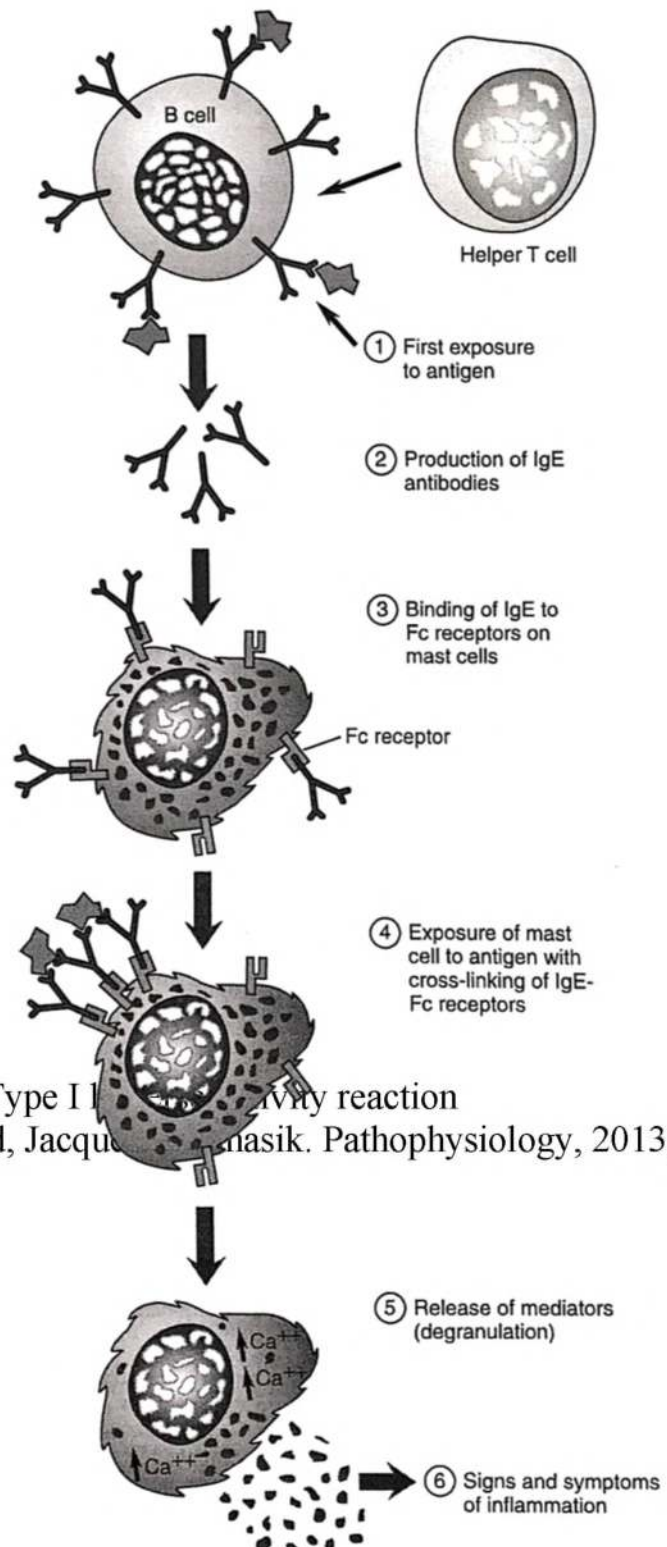


Figure 1. Type I Hypersensitivity reaction
(from Lee-Ellen Copstead, Jacquelyn Nasik. Pathophysiology, 2013)

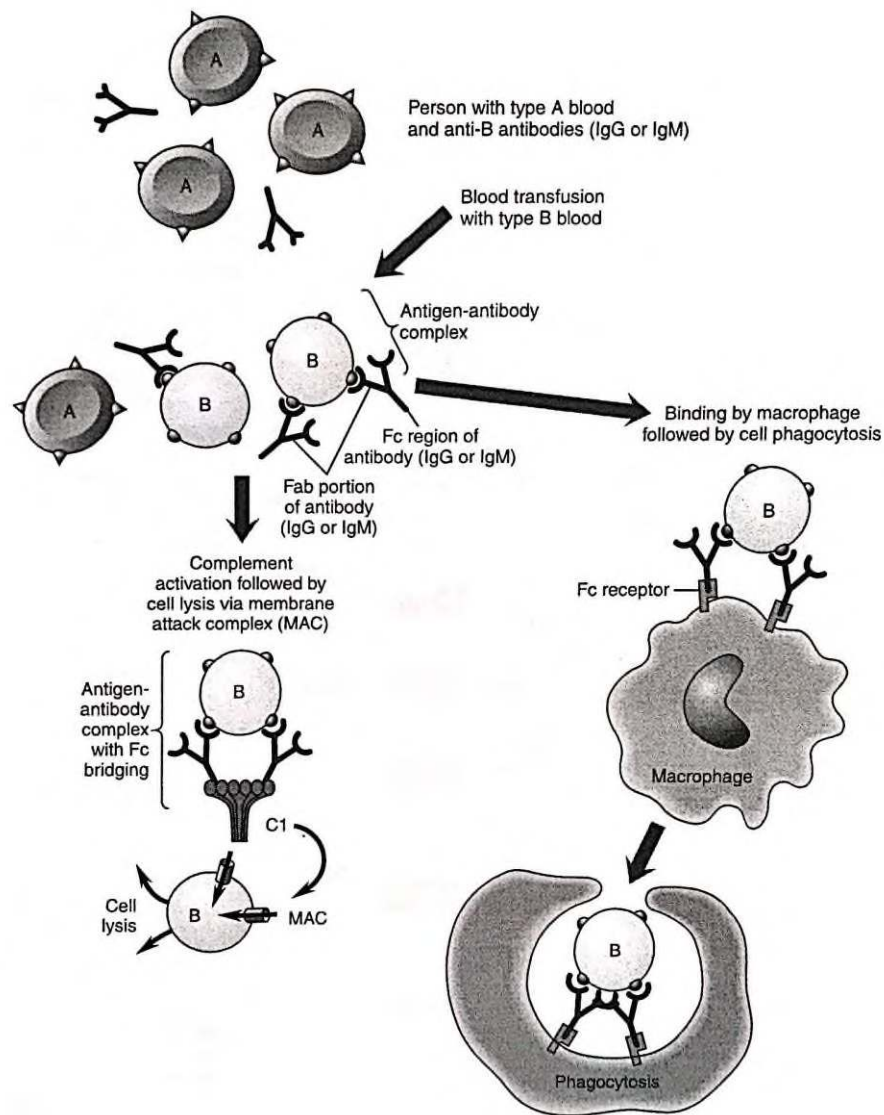


Figure 2. Type II hypersensitivity reaction.
(from Lee-Ellen Copstead, Jacquelyn Banasik. Pathophysiology, 2013).

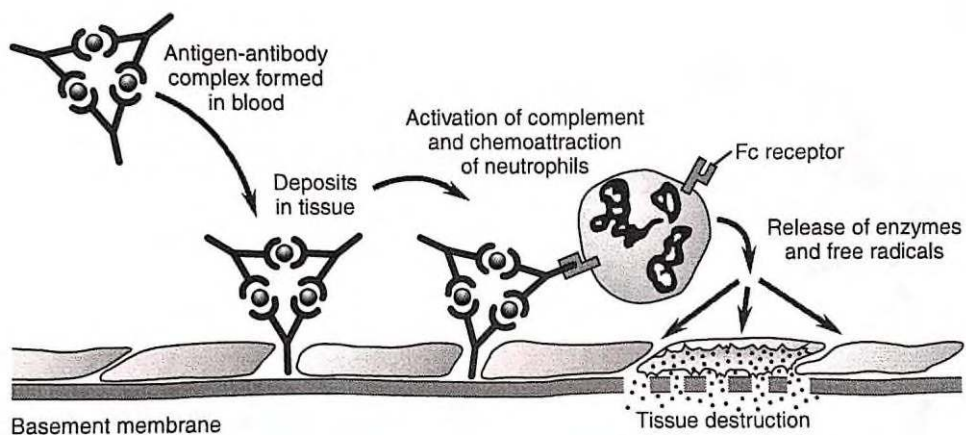


Figure 3. Type III hypersensitivity reaction
(from Lee-Ellen Copstead, Jacquelyn Banasik. Pathophysiology, 2013)

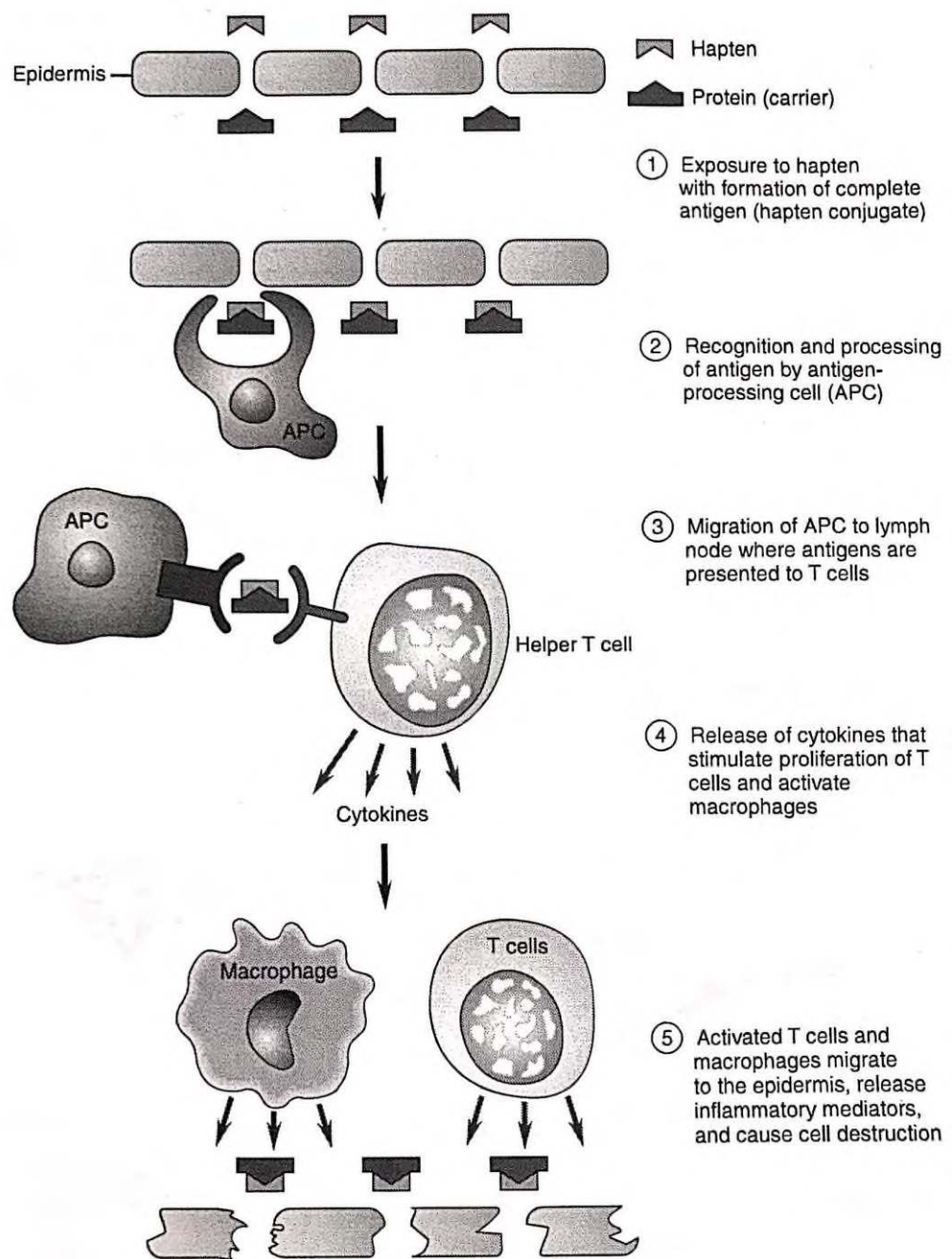


Figure 4. Type IV hypersensitivity reaction.
(from Lee-Ellen Copstead, Jacquelyn Banasik. Pathophysiology, 2013).

Theme: Inflammation

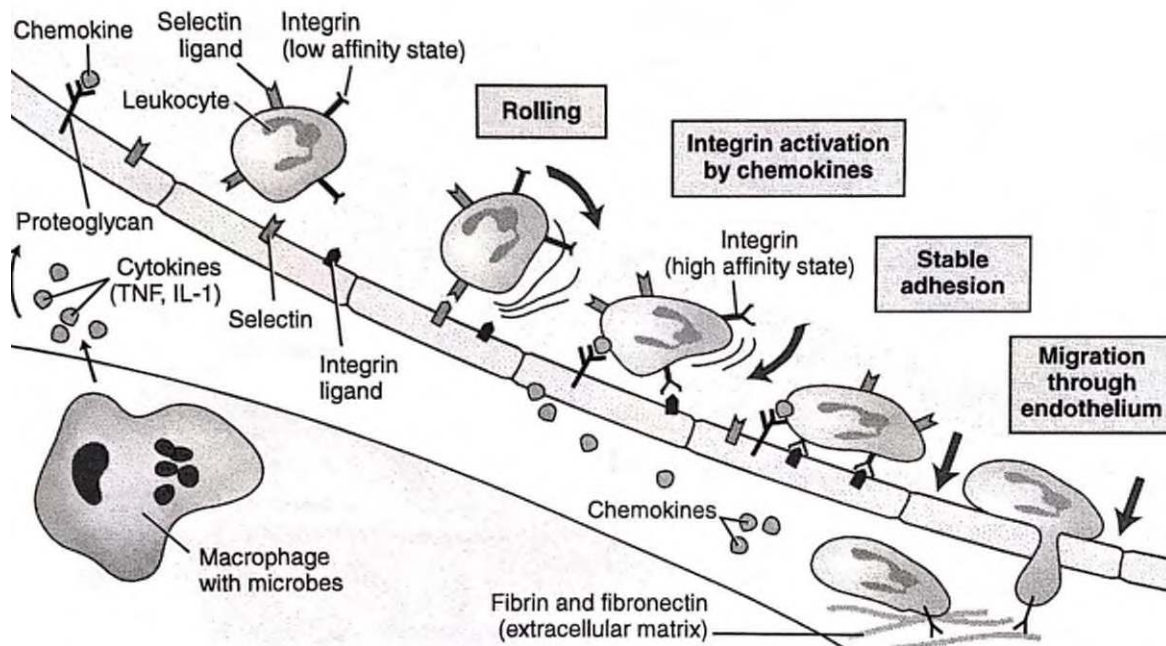
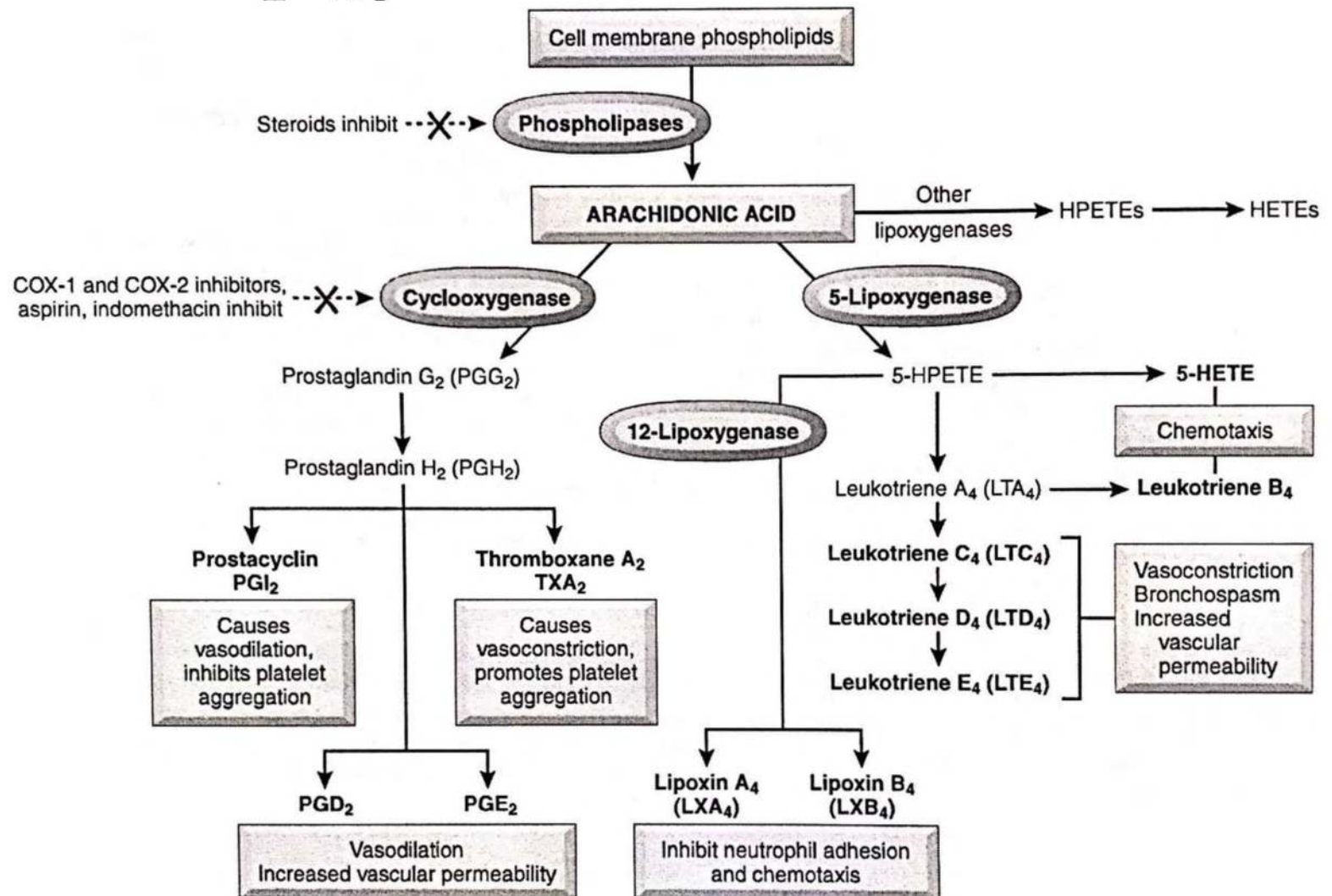


Figure 5. Emigration of neutrophils from the bloodstream into tissue is mediated by receptor interactions with the capillary endothelium. With inflammation and injury, endothelial cells began to express binding molecules on their cell surfaces (selectins). Leukocytes also have selectins, which can bind to endothelial adhesion proteins. The selectin interactions cause the leukocytes to stick and roll. Chemokines on the surface of endothelial cells interact with neutrophils (and macrophages) to increase the binding affinity of integrin receptors on leukocytes. Firm attachment and diapedesis through the capillary wall is facilitated by integrins, which allow the neutrophils to bind to endothelial cells and extracellular matrix and then pull themselves into the tissue. *IL*, interleukin; *TNF*, tumor necrosis factor.

(from Lee-Ellen Copstead, Jacquelyn Banasik. Pathophysiology, 2013).

Figure 6. Gen
arachidonic aci
(from Lee-1



Theme: Typical metabolic disorders of acid-base balance

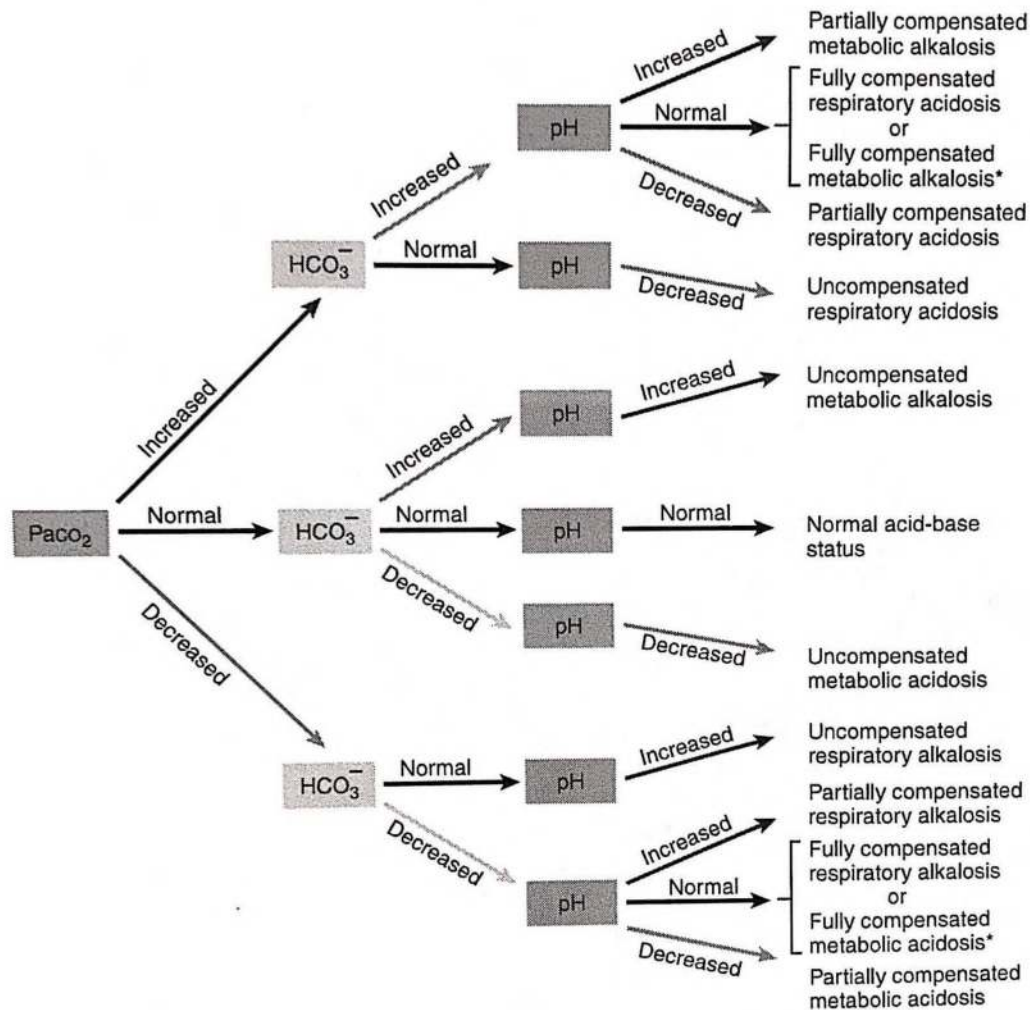


Figure 7. Flowchart for interpretation of laboratory measurements specific for acid – base imbalances. Use This flowchart to determine the primary acid – base from a set of laboratory values. Began on the left with PaCO₂ and follow the arrows. The flowchart does not include mixed acid – base imbalances.

(from Lee-Ellen Copstead, Jacquelyn Banasik. Pathophysiology, 2013).

TABLE 1.

Respiratory responses to changes in carbonic and metabolic acids

Stimulus	Respiratory response	Result
Increased Pa CO ₂ , Decreased pH	Hyperventilation	Correction of imbalance
Decreased Pa CO ₂ Increased pH	Hypoventilation	Correction of imbalance
Decreased pH from excess of metabolic acids	Hyperventilation	Compensation for imbalance
Increased pH from deficit of metabolic acids	Hypoventilation	Compensation for imbalance

TABLE 2.

Renal responses to changes in metabolic and carbonic acids

Stimulus	Renal response	Result
Decreased pH from excess of metabolic acids	Secrete more H ⁺ into renal tubules	Correction of imbalance
Increased pH from deficit of metabolic acids	Make more ammonia Secrete fewer H ⁺ into renal tubules Excrete HCO ₃ ⁻ Make less ammonia	Correction of imbalance
Decreased pH from excess of carbonic acid	Secrete more H ⁺ into renal tubules	Compensation for imbalance
Increased pH from deficit of carbonic acid	Excrete HCO ₃ ⁻ Make less ammonia	Compensation for imbalance

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